

Power Analysis for Confirmatory Factor Analysis: A Practical Tutorial

Analytical and Simulation-Based Approaches

Pablo Rogers

2026-04-10

This notebook presents a pedagogical tutorial on statistical power analysis in Confirmatory Factor Analysis (CFA). Proper planning of sample sizes requires assessing the power to detect overall model misfit by analytical approach or simulation-based approach. Standard analytical tools typically rely on asymptotic normal theory, which may break down when confronted with non-normality or missing data. This tutorial demonstrates both analytical strategies (using `semPower` and `semTools`) and Monte Carlo simulations (using `simsem`) side-by-side using the WHOQOL-BREF 4-factor model as a unifying case study. We illustrate how simulation can complement analytical calculations, allowing researchers to evaluate parameter-level power, coverage, and relative bias under realistic scenarios involving varying degrees of model misspecification, missing data, and continuous non-normal indicators. Ultimately, we argue that analytical methods are ideal for initial planning, while simulations constitute the gold standard for robust confirmation.

1 Introduction

Statistical power in Confirmatory Factor Analysis (CFA) reflects the probability of correctly rejecting an incorrectly specified model, or detecting a parameter of interest when it is truly non-zero in the population (MacCallum et al., 1996). Power analysis is critical when planning a CFA study; however, researchers often rely on arbitrary rules of thumb mapping required sample sizes to the number of measured variables or parameters, rather than formal prospective calculations.

There are two main paradigms to power analysis in CFA:

1. **Analytical methods:** Rely on asymptotic statistical theory (e.g., the non-central χ^2 distribution) to calculate the required sample size for detecting global model misfit based on established fit indices like the RMSEA (Moshagen & Bader, 2024).

2. **Simulation-based methods:** Use Monte Carlo data generation to repeatedly establish how the model behaves under assumed population models, allowing the estimation of power, bias, and coverage for specific parameters or overall fit indices across varying sample sizes (Pornprasertmanit et al., 2021).

This document provides a comprehensive tutorial covering both analytical methods using the `semPower` and `semTools` packages, and Monte Carlo methods using the `simsem` package.

Reproducibility Note

For local development and computational convenience across this pedagogical document, simulations use `N_REP = 50`. For production-quality applied research, you must increase this to `N_REP = 1,000`. The results throughout this tutorial will thus show more sampling variability than you would expect in a final study. All analytical and simulation parameters are centrally defined in `helpers.R` and `models.R`.

2 The Running Example: WHOQOL-BREF Population CFA Model

To anchor our power analyses, we use the WHOQOL-BREF measurement instrument. Meaningful power analysis requires the definition of a population model—the “true” state of the world from which we believe our data are generated. By simulating from a meta-analytically derived population model rather than a purely arbitrary one, researchers can ensure that tests reflect realistic boundaries (Lin & Yao, 2022).

We specify four population variations to perform sensitivity analyses:

1. **Meta-analytic:** Loadings and residual variances closely matching Lin & Yao (2022).
2. **Naive:** A pessimistic baseline where factor loadings are based on the lowest reported reliability values per domain (Mosqueira Taipe et al., 2026).
3. **Optimistic:** A best-case scenario where factor loadings are based on the highest reported reliability values per domain (Mosqueira Taipe et al., 2026).
4. **H1:** A misspecified model with cross-loadings based on the meta-EFA results from Lin & Yao (2022).

2.1 Population model

The primary population model represents a meta-analytic scenario based on Lin & Yao (2022). Factor loadings (λ) match the pooled empirical Exploratory Factor Analysis (EFA) estimates across 16 international studies. Characteristics:

- **Interfactor correlations (ϕ):** Set to 0.20 (conservative midpoint of the reported range 0.08–0.31).

- **Residual covariance:** $Q3$ and $Q4$ share a residual covariance ($r(Q3, Q4|\text{physical}) = 0.30$), maintaining the subfactor signal observed in the EFA 5-factor solution (Lin & Yao, 2022).
- **Factor structure:** Item $Q8$ is maintained in the environment domain despite having a higher cross-loading on the psychological domain in the meta-EFA ($\lambda_{\text{psy}} > \lambda_{\text{env}}$). This represents the null hypothesis (H_0) version of the population, strictly maintaining the original 4-factor structure without cross-loadings.

```
# Factor loadings: empirical estimates from Lin & Yao (2022)
psycho      =~ 0.92*Q5 + 0.81*Q6 + 0.94*Q7 +
              0.73*Q11 + 0.75*Q19 + 0.63*Q26

physical    =~ 0.84*Q3 + 0.83*Q4 + 0.63*Q10 +
              0.71*Q15 + 0.68*Q16 + 0.89*Q17 + 0.61*Q18

social      =~ 0.94*Q20 + 0.89*Q21 + 0.78*Q22

environment =~ 0.37*Q8 + 0.49*Q9 + 0.79*Q12 + 0.81*Q13 +
              0.77*Q14 + 0.75*Q23 + 0.64*Q24 + 0.73*Q25

# Residual covariance Q3~~Q4
# r(Q3,Q4|physical) = 0.30 (method bias / conceptual proximity)
# cov = 0.30 * sqrt(1-0.842) * sqrt(1-0.832) = 0.0908
Q4 ~~ 0.0908*Q3

# Factor variances (=1) and interfactor correlations
# Range reported in Lin & Yao (2022): 0.08 to 0.31
# Conservative midpoint = 0.20 for all pairs
psycho      ~~ 1*psycho      + 0.2*physical + 0.2*social + 0.2*environment
physical    ~~ 1*physical    + 0.2*social   + 0.2*environment
social      ~~ 1*social      + 0.2*environment
environment ~~ 1*environment

# Residual variances: = 12 (standardised indicators)
Q5 ~~ 0.1536*Q5 # 1 0.922
Q6 ~~ 0.3439*Q6 # 1 0.812
Q7 ~~ 0.1164*Q7 # 1 0.942
Q11 ~~ 0.4671*Q11 # 1 0.732
Q19 ~~ 0.4375*Q19 # 1 0.752
Q26 ~~ 0.6031*Q26 # 1 0.632
Q3  ~~ 0.2944*Q3  # 1 0.842
```

Q4	~~	0.3111*Q4	#	1	0.83 ²	
Q10	~~	0.6031*Q10	#	1	0.63 ²	
Q15	~~	0.4959*Q15	#	1	0.71 ²	
Q16	~~	0.5376*Q16	#	1	0.68 ²	
Q17	~~	0.2079*Q17	#	1	0.89 ²	
Q18	~~	0.6279*Q18	#	1	0.61 ²	
Q20	~~	0.1164*Q20	#	1	0.94 ²	
Q21	~~	0.2079*Q21	#	1	0.89 ²	
Q22	~~	0.3916*Q22	#	1	0.78 ²	
Q8	~~	0.8631*Q8	#	1	0.37 ²	- lowest loading; bridge item (betweenness=40.5)
Q9	~~	0.7599*Q9	#	1	0.49 ²	
Q12	~~	0.3759*Q12	#	1	0.79 ²	
Q13	~~	0.3439*Q13	#	1	0.81 ²	
Q14	~~	0.4071*Q14	#	1	0.77 ²	
Q23	~~	0.4375*Q23	#	1	0.75 ²	
Q24	~~	0.5904*Q24	#	1	0.64 ²	
Q25	~~	0.4671*Q25	#	1	0.73 ²	

Source: [Population Models](#)

2.2 Naive model

This model defines a pessimistic, empirically grounded scenario. We apply the Spearman-Brown prediction formula to derive tau-equivalent factor loadings based on the lowest reported Cronbach's alpha (α) values.

- **Approach:** Assuming equal loadings within each domain, we estimate the loadings as $\lambda = \sqrt{\alpha/(k - \alpha(k - 1))}$ and residual variances as $\theta = 1 - \lambda^2$.
- **Alpha sources:** Based on the lowest alphas for the original 4-factor structure reported by a single study identified in the systematic review by Mosqueira Taipe et al. (2026) (Physical $\alpha = 0.672$; Psychological $\alpha = 0.704$; Social $\alpha = 0.630$; Environmental $\alpha = 0.709$). All values reside within the $M - 1SD$ range reported by Lin & Yao (2022).
- **Interfactor correlations (ϕ):** Fixed at 0.30 (the upper bound from Lin & Yao (2022)), which makes factors harder to distinguish and yields a conservative power estimate.
- **Derived parameters:** E.g., Physical ($\lambda = 0.4758, \theta = 0.7736$), Psychological ($\lambda = 0.5328, \theta = 0.7161$), Social ($\lambda = 0.6017, \theta = 0.6379$), Environmental ($\lambda = 0.4832, \theta = 0.7665$). The residual covariance between $Q3$ and $Q4$ is adjusted to 0.2321.

```
# Factor loadings: tau-equivalent (uniform per domain)
psycho      =~ 0.5328*Q5 + 0.5328*Q6 + 0.5328*Q7 +
              0.5328*Q11 + 0.5328*Q19 + 0.5328*Q26
```

```

physical    =~ 0.4758*Q3  + 0.4758*Q4  + 0.4758*Q10 +
              0.4758*Q15 + 0.4758*Q16 + 0.4758*Q17  + 0.4758*Q18

social      =~ 0.6017*Q20 + 0.6017*Q21 + 0.6017*Q22

environment =~ 0.4832*Q8  + 0.4832*Q9  + 0.4832*Q12 + 0.4832*Q13 +
              0.4832*Q14 + 0.4832*Q23  + 0.4832*Q24 + 0.4832*Q25

# Residual covariance Q3~~Q4
# r(Q3,Q4|physical) = 0.30; cov = 0.30 × (1 - 0.47582) = 0.2321
Q4 ~~ 0.2321*Q3

# Factor variances (=1) and interfactor correlations
# = 0.30: upper bound of Lin & Yao (2022) range - pessimistic scenario
psycho      ~~ 1*psycho      + 0.3*physical + 0.3*social + 0.3*environment
physical    ~~ 1*physical    + 0.3*social  + 0.3*environment
social      ~~ 1*social      + 0.3*environment
environment ~~ 1*environment

# Residual variances: = 1 - 0.32 per domain
Q5  ~~ 0.7161*Q5    # Psychological: = 1 - 0.53282 = 0.7161
Q6  ~~ 0.7161*Q6
Q7  ~~ 0.7161*Q7
Q11 ~~ 0.7161*Q11
Q19 ~~ 0.7161*Q19
Q26 ~~ 0.7161*Q26
Q3  ~~ 0.7736*Q3    # Physical: = 1 - 0.47582 = 0.7736
Q4  ~~ 0.7736*Q4
Q10 ~~ 0.7736*Q10
Q15 ~~ 0.7736*Q15
Q16 ~~ 0.7736*Q16
Q17 ~~ 0.7736*Q17
Q18 ~~ 0.7736*Q18
Q20 ~~ 0.6379*Q20   # Social: = 1 - 0.60172 = 0.6379
Q21 ~~ 0.6379*Q21
Q22 ~~ 0.6379*Q22
Q8  ~~ 0.7665*Q8    # Environmental: = 1 - 0.48322 = 0.7665
Q9  ~~ 0.7665*Q9
Q12 ~~ 0.7665*Q12
Q13 ~~ 0.7665*Q13
Q14 ~~ 0.7665*Q14
Q23 ~~ 0.7665*Q23

```

Q24 ~~ 0.7665*Q24
Q25 ~~ 0.7665*Q25

Source: [Population Models](#)

2.3 Optimistic model

This model defines an optimistic, empirically grounded scenario using the same Spearman-Brown derivation approach but leveraging the highest observed Cronbach's alpha (α) values per domain.

- **Alpha sources:** Based on maximum alphas reported in the literature (Mosqueira Taípe et al., 2026) (Physical $\alpha = 0.860$; Psychological $\alpha = 0.860$; Social $\alpha = 0.810$; Environmental $\alpha = 0.849$). All values are confirmed within the $M + 1SD$ range from Lin & Yao (2022).
- **Interfactor correlations (ϕ):** Fixed at 0.08 (the lower bound from Lin & Yao (2022)), resulting in factors that are well-distinguished and providing an optimistic power estimate.
- **Derived parameters:** E.g., Physical ($\lambda = 0.6837, \theta = 0.5326$), Psychological ($\lambda = 0.7113, \theta = 0.4941$), Social ($\lambda = 0.7661, \theta = 0.4130$), Environmental ($\lambda = 0.6424, \theta = 0.5873$). The residual covariance between $Q3$ and $Q4$ is adjusted to 0.1598.

```
# Factor loadings: tau-equivalent (uniform per domain)
psycho      =~ 0.7113*Q5  + 0.7113*Q6  + 0.7113*Q7  +
              0.7113*Q11 + 0.7113*Q19 + 0.7113*Q26

physical    =~ 0.6837*Q3  + 0.6837*Q4  + 0.6837*Q10 +
              0.6837*Q15 + 0.6837*Q16 + 0.6837*Q17  + 0.6837*Q18

social      =~ 0.7661*Q20 + 0.7661*Q21 + 0.7661*Q22

environment =~ 0.6424*Q8  + 0.6424*Q9  + 0.6424*Q12 + 0.6424*Q13 +
              0.6424*Q14 + 0.6424*Q23  + 0.6424*Q24 + 0.6424*Q25

# Residual covariance Q3~~Q4
# r(Q3,Q4|physical) = 0.30; cov = 0.30 × (1 - 0.68372) = 0.1598
Q4 ~~ 0.1598*Q3

# Factor variances (=1) and interfactor correlations
# = 0.08: lower bound of Lin & Yao (2022) range - optimistic scenario
psycho      ~~ 1*psycho      + 0.08*physical + 0.08*social + 0.08*environment
physical    ~~ 1*physical    + 0.08*social   + 0.08*environment
social      ~~ 1*social      + 0.08*environment
```

```

environment ~~ 1*environment

# Residual variances: = 12 per domain
Q5 ~~ 0.4941*Q5      # Psychological: = 1 0.71132 = 0.4941
Q6 ~~ 0.4941*Q6
Q7 ~~ 0.4941*Q7
Q11 ~~ 0.4941*Q11
Q19 ~~ 0.4941*Q19
Q26 ~~ 0.4941*Q26
Q3  ~~ 0.5326*Q3      # Physical: = 1 0.68372 = 0.5326
Q4  ~~ 0.5326*Q4
Q10 ~~ 0.5326*Q10
Q15 ~~ 0.5326*Q15
Q16 ~~ 0.5326*Q16
Q17 ~~ 0.5326*Q17
Q18 ~~ 0.5326*Q18
Q20 ~~ 0.4130*Q20     # Social: = 1 0.76612 = 0.4130
Q21 ~~ 0.4130*Q21
Q22 ~~ 0.4130*Q22
Q8  ~~ 0.5873*Q8      # Environmental: = 1 0.64242 = 0.5873
Q9  ~~ 0.5873*Q9
Q12 ~~ 0.5873*Q12
Q13 ~~ 0.5873*Q13
Q14 ~~ 0.5873*Q14
Q23 ~~ 0.5873*Q23
Q24 ~~ 0.5873*Q24
Q25 ~~ 0.5873*Q25

```

Source: [Population Models](#)

2.4 Misspecification model

This represents the alternative hypothesis (H_1), defining a cross-loading population model. Three empirical cross-loadings are added to the baseline 4-factor structure based on Lin & Yao (2022) and supported by recent literature (Mosqueira Taípe et al., 2026):

- **Cross-loading 1:** item Q_8 loading on the Psychological factor ($\lambda = 0.55$). This item showed the highest betweenness (40.50) and explicitly grouped with the psychological factor in 9/16 studies.
- **Cross-loading 2:** item Q_9 loading on the Psychological factor ($\lambda = 0.21$). Identified as a secondary loading in the 5-factor solution.

- **Cross-loading 3:** item *Q15* loading on the Environmental factor ($\lambda = 0.24$). “Mobility” conceptually bridges physical ability and environmental access, showing the second-highest betweenness (36.50).

Residual variances (θ) for cross-loaded items were recalculated to preserve total item variance using the formula $\theta = 1 - \lambda_{\text{primary}}^2 - \lambda_{\text{secondary}}^2 - 2\lambda_{\text{primary}}\lambda_{\text{secondary}}\phi$, where the interfactor correlation (ϕ) is 0.20.

```
# Factor loadings (with three cross-loadings)
psycho      =~ 0.92*Q5  + 0.81*Q6  + 0.94*Q7  +
               0.73*Q11 + 0.75*Q19 + 0.63*Q26 +
               0.55*Q8  +                                     # cross-load (1)
               0.21*Q9                                     # cross-load (2)

physical    =~ 0.84*Q3  + 0.83*Q4  + 0.63*Q10 +
               0.71*Q15 + 0.68*Q16 + 0.89*Q17 + 0.61*Q18

social      =~ 0.94*Q20 + 0.89*Q21 + 0.78*Q22

environment =~ 0.37*Q8  + 0.49*Q9  + 0.79*Q12 + 0.81*Q13 +
               0.77*Q14 + 0.75*Q23  + 0.64*Q24 + 0.73*Q25 +
               0.24*Q15                                     # cross-load (3)

# Residual covariance Q3~~Q4 (fixed in both P2 and analyzeModel)
Q4 ~~ 0.0908*Q3

# Factor variances (=1) and interfactor correlations
psycho      ~~ 1*psycho      + 0.2*physical + 0.2*social + 0.2*environment
physical    ~~ 1*physical    + 0.2*social   + 0.2*environment
social      ~~ 1*social      + 0.2*environment
environment ~~ 1*environment

# Residual variances
Q5  ~~ 0.1536*Q5
Q6  ~~ 0.3439*Q6
Q7  ~~ 0.1164*Q7
Q11 ~~ 0.4671*Q11
Q19 ~~ 0.4375*Q19
Q26 ~~ 0.6031*Q26
Q3   ~~ 0.2944*Q3
Q4   ~~ 0.3111*Q4
Q10 ~~ 0.6031*Q10
```

```

Q15 ~~ 0.3701*Q15    # UPDATED: was 0.4959 - cross-load on environment
Q16 ~~ 0.5376*Q16
Q17 ~~ 0.2079*Q17
Q18 ~~ 0.6279*Q18
Q20 ~~ 0.1164*Q20
Q21 ~~ 0.2079*Q21
Q22 ~~ 0.3916*Q22
Q8  ~~ 0.4792*Q8     # UPDATED: was 0.8631 - cross-load on psycho
Q9  ~~ 0.6746*Q9     # UPDATED: was 0.7599 - cross-load on psycho
Q12 ~~ 0.3759*Q12
Q13 ~~ 0.3439*Q13
Q14 ~~ 0.4071*Q14
Q23 ~~ 0.4375*Q23
Q24 ~~ 0.5904*Q24
Q25 ~~ 0.4671*Q25

```

Source: [Population Models](#)

2.5 Comparison table

Below is a summary of the four population models. Note that the H1 model is the misspecification model.

Table 1: Summary of population model variants used for sensitivity analysis

Model	Loading Range	Mean Loading	Factor Correlations	Residual Range	Q3–Q4 Correlation
Meta-analytic (popModel)	0.37–0.94	0.74	0.20	0.12–0.86	0.09
Naive (naiveModel)	0.48–0.60	0.51	0.30	0.64–0.77	0.23
Optimistic (optModel)	0.64–0.77	0.69	0.08	0.41–0.59	0.16
H1 (H1model)	0.21–0.94	0.78	0.20	0.12–0.67	0.09

Source: [Population Models](#)

2.6 Analysis model

The analysis model serves dual purposes depending on the simulation paradigm:

- **Paradigm A (Recovery power):** Estimates the 4-factor structure plus the $Q3$ and $Q4$ residual covariance given data from the meta-analytic, naive, or optimistic population models.
- **Paradigm B (Misspecification detection):** Acts as the null hypothesis (H_0), meaning it *omits* the cross-loadings, when fitting data generated from the misspecification model (H_1). Here, power is the probability of correctly rejecting the model given that cross-loadings exist in the population.

Key constraints and features:

- **Identification:** Factor variances are fixed to 1 (marker-free scaling) allowing the first indicator to be freely estimated.
- **Residual covariance:** $Q3$ and $Q4$ correlation is freely estimated, which is theoretically justified and empirically supported (Lin & Yao, 2022; Mosqueira Taípe et al., 2026).
- **Cross-loadings:** The omission of cross-loadings for items $Q8$, $Q9$, and $Q15$ deliberately defines the misspecification tested in Paradigm B.

```
# Factor loadings: freely estimated
psycho      =~ Q5  + Q6  + Q7  + Q11 + Q19 + Q26
physical    =~ Q3  + Q4  + Q10 + Q15 + Q16 + Q17 + Q18
social      =~ Q20 + Q21 + Q22
environment =~ Q8  + Q9  + Q12 + Q13 + Q14 + Q23 + Q24 + Q25

# Factor variances (=1) and free interfactor correlations
psycho      ~~ 1*psycho      + physical + social + environment
physical    ~~ 1*physical    + social    + environment
social      ~~ 1*social      + environment
environment ~~ 1*environment

# Residual covariance Q3~~Q4: freely estimated
Q4 ~~ Q3

# Residual variances: freely estimated
Q5 ~~ Q5; Q6 ~~ Q6; Q7 ~~ Q7
Q11 ~~ Q11; Q19 ~~ Q19; Q26 ~~ Q26
Q3 ~~ Q3; Q4 ~~ Q4; Q10 ~~ Q10
Q15 ~~ Q15; Q16 ~~ Q16; Q17 ~~ Q17; Q18 ~~ Q18
Q20 ~~ Q20; Q21 ~~ Q21; Q22 ~~ Q22
Q8  ~~ Q8; Q9  ~~ Q9; Q12 ~~ Q12; Q13 ~~ Q13
Q14 ~~ Q14; Q23 ~~ Q23; Q24 ~~ Q24; Q25 ~~ Q25
```

Source: [Population Models](#)

3 Analytical Power Analysis

Analytical power usually asks: “How large must N be to reject the hypothesis that the model fits perfectly (or adequately) when the true misfit in the population is of a certain magnitude?” A common formulation sets the alternative hypothesis to $RMSEA = 0.05$ (MacCallum et al., 1996).

3.1 Model Degrees of Freedom

Analytical calculations depend crucially on model degrees of freedom (df), the difference between the number of available pieces of information (unique variances/covariances) and the number of freely estimated parameters. Knowing this allows tools to parameterize the non-central χ^2 distribution. This calculation is necessary for both a priori and post hoc power analyses in analytical methods for global fit.

Model degrees of freedom (via `semPower`): 249

Source: [Analytical Power Analysis](#)

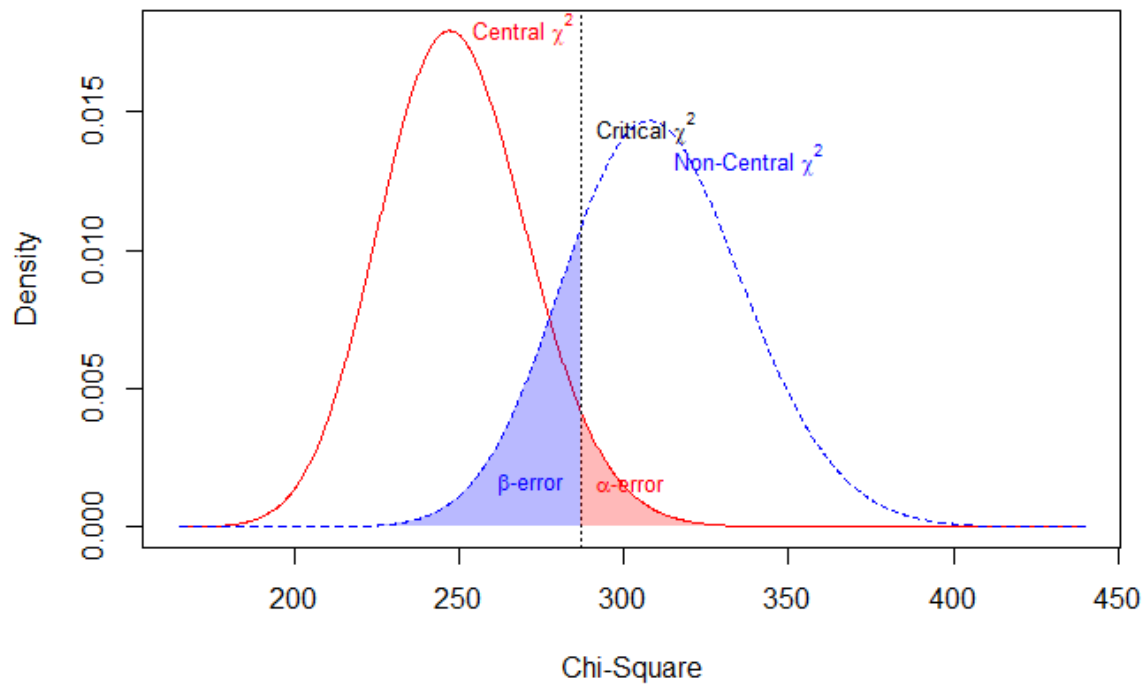
3.2 A Priori: How Many Observations?

```
semPower: A priori power analysis

FO                0.622500
RMSEA             0.050000
Mc               0.732531
GFI              0.950683
AGFI             0.940582

df                249
Required Num Observations 99

Critical Chi-Square 286.8077
NCP               61.00500
Alpha            0.050000
Beta            0.199217
Power (1 - Beta) 0.800783
Implied Alpha/Beta Ratio 0.250983
```



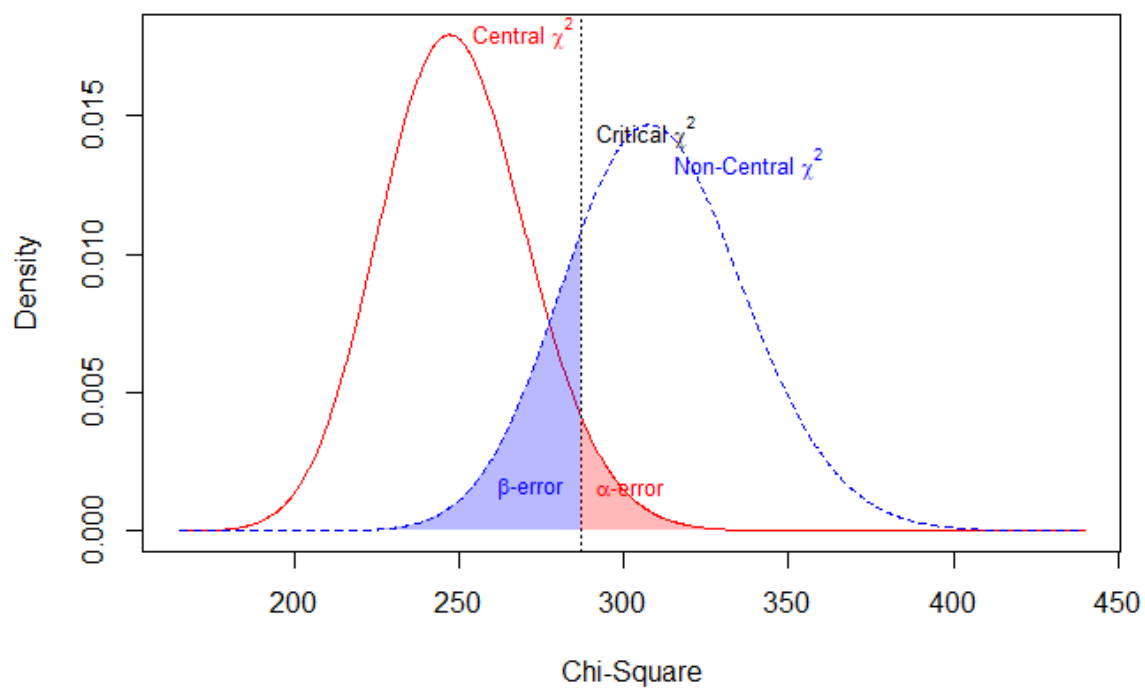
Source: [Analytical Power Analysis](#)

semPower: A priori power analysis

F0	0.303870
RMSEA	0.034934
Mc	0.859044

df	249
Required Num Observations	202

Critical Chi-Square	286.8077
NCP	61.07791
Alpha	0.050000
Beta	0.198481
Power (1 - Beta)	0.801519
Implied Alpha/Beta Ratio	0.251914



Source: [Analytical Power Analysis](#)

3.3 Power Curves

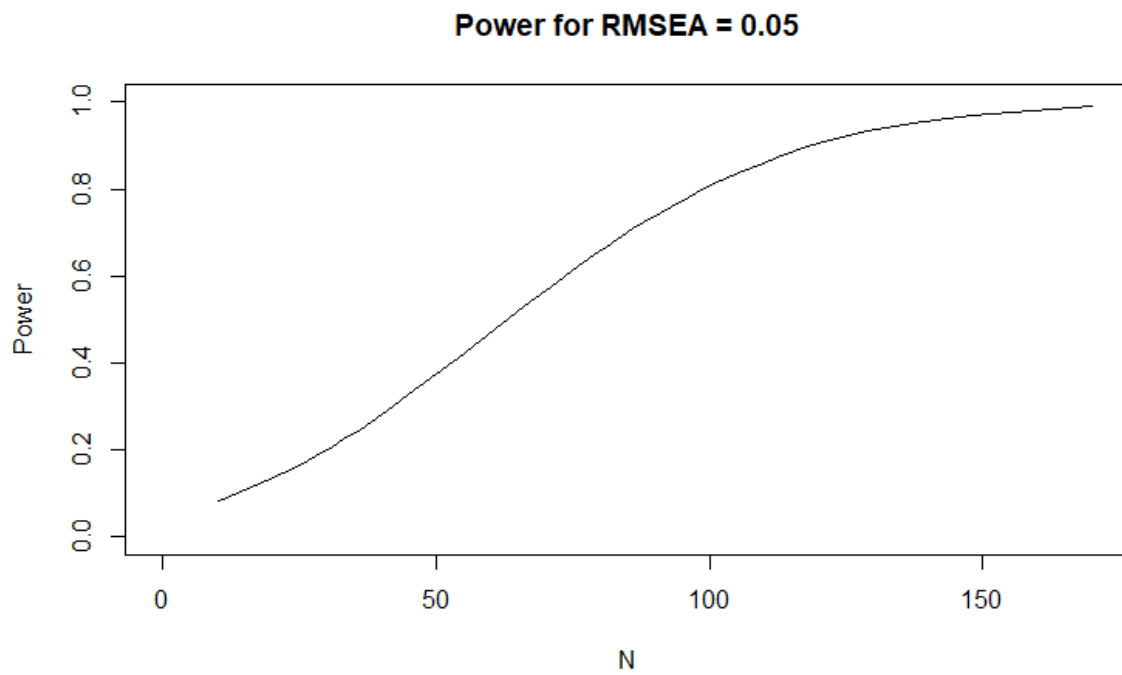


Figure 1: Power as a function of sample size for RMSEA = 0.05

Source: [Analytical Power Analysis](#)

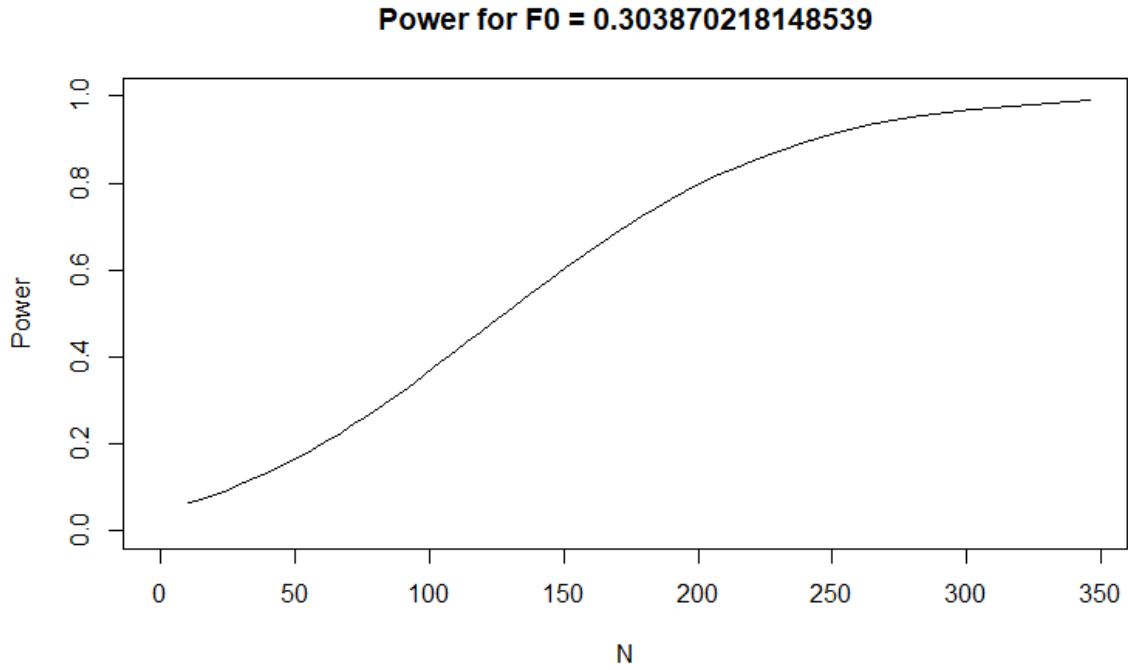


Figure 2: Power as a function of sample size for the population model (popModel)

Source: [Analytical Power Analysis](#)

Table 2: Summary of analytical power analysis results

Method	N Required	Power	Implied RMSEA	Implied Fmin	df
Model-free a priori (RMSEA .05)	99	0.8008	0.0500	0.6225	249
Model-based a priori (Population Model)	202	0.8015	0.0349	0.3039	249

Source: [Analytical Power Analysis](#)

! Limitations of the Analytical Approach

The RMSEA/ χ^2 -based analytical method classically assumes continuous, multivariate normal data and Maximum Likelihood estimation. It can yield severely biased optimal sample sizes when distributions are non-normal, data are ordinal, or missingness induces

efficiency losses. Furthermore, it measures power regarding *global misfit*, telling us nothing about whether N is adequate to detect a specific weak loading or structural path.

4 Monte Carlo Simulation: Ideal Conditions

Given the limitations above, Monte Carlo simulation stands as a highly flexible strategy. We generate many synthetic datasets from the defined `popModel`, apply our `analyzeModel`, and record the parameter estimates and standard errors.

Under strictly ideal conditions (multivariate normal continuous data, no missing values), simulation results should match analytical calculations, validating both implementations. Notice that varying sample sizes gives us empirical power curves.

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	298.104	5878.811	6042.462	0.038	0.975	0.972	0.065
2	225	292.468	12628.755	12813.149	0.031	0.981	0.979	0.055
3	350	286.833	19378.698	19583.836	0.024	0.987	0.985	0.046
4	475	281.197	26128.641	26354.523	0.017	0.993	0.992	0.036
5	600	275.561	32878.585	33125.209	0.010	0.999	0.999	0.026

===== Parameter Estimates and Standard Errors =====

	Estimate	Average	Estimate	SD	Average	SE
psycho=~Q5	0.918		0.049		0.045	
psycho=~Q6	0.819		0.054		0.049	
psycho=~Q7	0.944		0.048		0.044	
psycho=~Q11	0.742		0.057		0.051	
psycho=~Q19	0.749		0.044		0.050	
psycho=~Q26	0.630		0.060		0.053	
physical=~Q3	0.827		0.055		0.048	
physical=~Q4	0.827		0.057		0.049	
physical=~Q10	0.625		0.057		0.054	
physical=~Q15	0.699		0.060		0.053	
physical=~Q16	0.681		0.053		0.052	
physical=~Q17	0.894		0.053		0.047	
physical=~Q18	0.611		0.050		0.054	
social=~Q20	0.930		0.052		0.045	
social=~Q21	0.886		0.050		0.047	
social=~Q22	0.768		0.043		0.050	

environment=~Q8	0.369	0.056	0.059
environment=~Q9	0.491	0.061	0.057
environment=~Q12	0.787	0.055	0.050
environment=~Q13	0.800	0.056	0.050
environment=~Q14	0.772	0.046	0.051
environment=~Q23	0.747	0.047	0.052
environment=~Q24	0.626	0.064	0.054
environment=~Q25	0.737	0.047	0.052
psycho~~physical	0.211	0.063	0.060
psycho~~social	0.209	0.060	0.059
psycho~~environment	0.217	0.071	0.060
physical~~social	0.204	0.061	0.060
physical~~environment	0.194	0.069	0.062
social~~environment	0.214	0.061	0.061
Q3~~Q4	0.087	0.026	0.026
Q5~~Q5	0.154	0.018	0.019
Q6~~Q6	0.342	0.036	0.032
Q7~~Q7	0.118	0.020	0.017
Q11~~Q11	0.470	0.038	0.041
Q19~~Q19	0.437	0.032	0.039
Q26~~Q26	0.605	0.039	0.052
Q3~~Q3	0.281	0.032	0.032
Q4~~Q4	0.313	0.036	0.035
Q10~~Q10	0.603	0.043	0.053
Q15~~Q15	0.510	0.052	0.046
Q16~~Q16	0.527	0.054	0.047
Q17~~Q17	0.202	0.025	0.027
Q18~~Q18	0.620	0.057	0.054
Q20~~Q20	0.117	0.032	0.028
Q21~~Q21	0.204	0.030	0.030
Q22~~Q22	0.392	0.032	0.037
Q8~~Q8	0.866	0.073	0.073
Q9~~Q9	0.758	0.070	0.065
Q12~~Q12	0.373	0.040	0.038
Q13~~Q13	0.342	0.038	0.036
Q14~~Q14	0.411	0.042	0.041
Q23~~Q23	0.435	0.039	0.042
Q24~~Q24	0.586	0.044	0.052
Q25~~Q25	0.458	0.042	0.044

	Power (Not equal 0)	Std Est	Std Est	SD	Std Ave SE
psycho=~Q5	1.000	0.919	0.013		0.012
psycho=~Q6	1.000	0.813	0.023		0.021
psycho=~Q7	1.000	0.940	0.010		0.010

psycho=~Q11	1.000	0.733	0.032	0.028
psycho=~Q19	1.000	0.749	0.023	0.027
psycho=~Q26	1.000	0.628	0.042	0.037
physical=~Q3	1.000	0.841	0.023	0.022
physical=~Q4	1.000	0.827	0.026	0.023
physical=~Q10	1.000	0.626	0.040	0.038
physical=~Q15	1.000	0.698	0.039	0.033
physical=~Q16	1.000	0.683	0.040	0.034
physical=~Q17	1.000	0.893	0.016	0.017
physical=~Q18	1.000	0.613	0.039	0.039
social=~Q20	1.000	0.938	0.018	0.016
social=~Q21	1.000	0.891	0.017	0.018
social=~Q22	1.000	0.774	0.025	0.026
environment=~Q8	1.000	0.368	0.053	0.053
environment=~Q9	1.000	0.490	0.054	0.047
environment=~Q12	1.000	0.789	0.028	0.026
environment=~Q13	1.000	0.806	0.024	0.025
environment=~Q14	1.000	0.769	0.025	0.028
environment=~Q23	1.000	0.749	0.022	0.029
environment=~Q24	1.000	0.631	0.039	0.039
environment=~Q25	1.000	0.736	0.029	0.030
psycho~~physical	0.941	0.211	0.063	0.060
psycho~~social	0.961	0.209	0.060	0.059
psycho~~environment	0.922	0.217	0.071	0.060
physical~~social	0.863	0.204	0.061	0.060
physical~~environment	0.824	0.194	0.069	0.062
social~~environment	0.961	0.214	0.061	0.061
Q3~~Q4	0.902	0.292	0.069	0.070
Q5~~Q5	1.000	0.155	0.024	0.022
Q6~~Q6	1.000	0.338	0.038	0.035
Q7~~Q7	1.000	0.117	0.020	0.019
Q11~~Q11	1.000	0.461	0.046	0.042
Q19~~Q19	1.000	0.438	0.034	0.041
Q26~~Q26	1.000	0.604	0.051	0.046
Q3~~Q3	1.000	0.292	0.038	0.036
Q4~~Q4	1.000	0.315	0.043	0.038
Q10~~Q10	1.000	0.607	0.049	0.048
Q15~~Q15	1.000	0.511	0.054	0.045
Q16~~Q16	1.000	0.532	0.055	0.046
Q17~~Q17	1.000	0.203	0.028	0.030
Q18~~Q18	1.000	0.623	0.048	0.048
Q20~~Q20	0.902	0.120	0.034	0.030
Q21~~Q21	1.000	0.206	0.030	0.033

Q22~~Q22	1.000	0.400	0.039	0.041
Q8~~Q8	1.000	0.862	0.039	0.039
Q9~~Q9	1.000	0.757	0.053	0.046
Q12~~Q12	1.000	0.377	0.044	0.041
Q13~~Q13	1.000	0.349	0.040	0.040
Q14~~Q14	1.000	0.408	0.038	0.043
Q23~~Q23	1.000	0.438	0.033	0.044
Q24~~Q24	1.000	0.600	0.049	0.048
Q25~~Q25	1.000	0.458	0.043	0.045
	Average Param	Average Bias	Coverage	r_coef.n r_se.n
psycho=~Q5	0.920	-0.002	0.922	-0.234 -0.934
psycho=~Q6	0.810	0.009	0.922	-0.337 -0.928
psycho=~Q7	0.940	0.004	0.902	-0.227 -0.935
psycho=~Q11	0.730	0.012	0.902	-0.414 -0.926
psycho=~Q19	0.750	-0.001	0.980	-0.190 -0.931
psycho=~Q26	0.630	0.000	0.922	-0.130 -0.936
physical=~Q3	0.840	-0.013	0.961	0.172 -0.919
physical=~Q4	0.830	-0.003	0.941	0.141 -0.919
physical=~Q10	0.630	-0.005	0.922	0.023 -0.935
physical=~Q15	0.710	-0.011	0.941	0.091 -0.913
physical=~Q16	0.680	0.001	0.961	-0.116 -0.941
physical=~Q17	0.890	0.004	0.941	-0.036 -0.913
physical=~Q18	0.610	0.001	0.941	-0.167 -0.938
social=~Q20	0.940	-0.010	0.902	-0.071 -0.938
social=~Q21	0.890	-0.004	0.961	-0.075 -0.919
social=~Q22	0.780	-0.012	1.000	0.203 -0.929
environment=~Q8	0.370	-0.001	0.980	0.002 -0.923
environment=~Q9	0.490	0.001	0.980	-0.098 -0.932
environment=~Q12	0.790	-0.003	0.980	-0.190 -0.925
environment=~Q13	0.810	-0.010	0.961	-0.242 -0.925
environment=~Q14	0.770	0.002	1.000	-0.244 -0.923
environment=~Q23	0.750	-0.003	0.941	-0.268 -0.927
environment=~Q24	0.640	-0.014	0.902	-0.020 -0.946
environment=~Q25	0.730	0.007	0.980	-0.179 -0.935
psycho~~physical	0.200	0.011	0.941	-0.059 -0.945
psycho~~social	0.200	0.009	0.941	-0.113 -0.944
psycho~~environment	0.200	0.017	0.902	-0.098 -0.929
physical~~social	0.200	0.004	0.922	-0.126 -0.938
physical~~environment	0.200	-0.006	0.902	-0.050 -0.939
social~~environment	0.200	0.014	0.922	-0.098 -0.932
Q3~~Q4	0.091	-0.003	0.980	-0.107 -0.901
Q5~~Q5	0.154	0.000	0.961	-0.213 -0.895
Q6~~Q6	0.344	-0.002	0.922	0.080 -0.861

Q7~~Q7	0.116	0.001	0.922	-0.226	-0.896
Q11~~Q11	0.467	0.003	0.941	0.055	-0.898
Q19~~Q19	0.438	0.000	0.980	0.006	-0.885
Q26~~Q26	0.603	0.002	0.980	0.068	-0.912
Q3~~Q3	0.294	-0.014	0.941	0.164	-0.879
Q4~~Q4	0.311	0.002	0.980	-0.261	-0.895
Q10~~Q10	0.603	0.000	0.980	0.069	-0.916
Q15~~Q15	0.496	0.015	0.961	-0.114	-0.853
Q16~~Q16	0.538	-0.011	0.922	0.270	-0.908
Q17~~Q17	0.208	-0.006	0.941	0.183	-0.915
Q18~~Q18	0.628	-0.008	0.902	-0.003	-0.903
Q20~~Q20	0.116	0.000	0.941	0.071	-0.909
Q21~~Q21	0.208	-0.004	0.961	-0.166	-0.884
Q22~~Q22	0.392	0.001	0.980	-0.097	-0.895
Q8~~Q8	0.863	0.003	0.980	0.075	-0.874
Q9~~Q9	0.760	-0.002	0.961	0.033	-0.882
Q12~~Q12	0.376	-0.002	0.980	-0.010	-0.875
Q13~~Q13	0.344	-0.002	0.922	0.022	-0.863
Q14~~Q14	0.407	0.004	0.941	0.095	-0.870
Q23~~Q23	0.438	-0.002	0.961	-0.166	-0.894
Q24~~Q24	0.590	-0.005	1.000	-0.035	-0.932
Q25~~Q25	0.467	-0.009	0.922	-0.120	-0.912

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.375	-0.375	0.885	-0.787	-0.895	0.550	-0.373
aic	-0.375	1.000	1.000	-0.649	0.642	0.540	-0.912	1.000
bic	-0.375	1.000	1.000	-0.650	0.643	0.540	-0.912	1.000
rmsea	0.885	-0.649	-0.650	1.000	-0.942	-0.950	0.785	-0.648
cfi	-0.787	0.642	0.643	-0.942	1.000	0.950	-0.774	0.640
tli	-0.895	0.540	0.540	-0.950	0.950	1.000	-0.731	0.537
srmr	0.550	-0.912	-0.912	0.785	-0.774	-0.731	1.000	-0.911
n	-0.373	1.000	1.000	-0.648	0.640	0.537	-0.911	1.000

===== Replications =====

Number of replications = 51

Number of converged replications = 51

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Ideal Conditions](#)

```

--- Power at N = 300 ---      cfi      tli      rmsea      srmr
"< 0.48" "< 0.42" "< 0.46" "< 0.42"
--- Power at N = 500 ---      cfi      tli      rmsea      srmr
"< 0.48" "< 0.48" "< 0.46" "< 0.42"
--- Power at N = 1000 ---     cfi      tli      rmsea      srmr
"< 0.42" "< 0.48" "< 0.46" "< 0.48"

```

Source: [Simulation — Ideal Conditions](#)

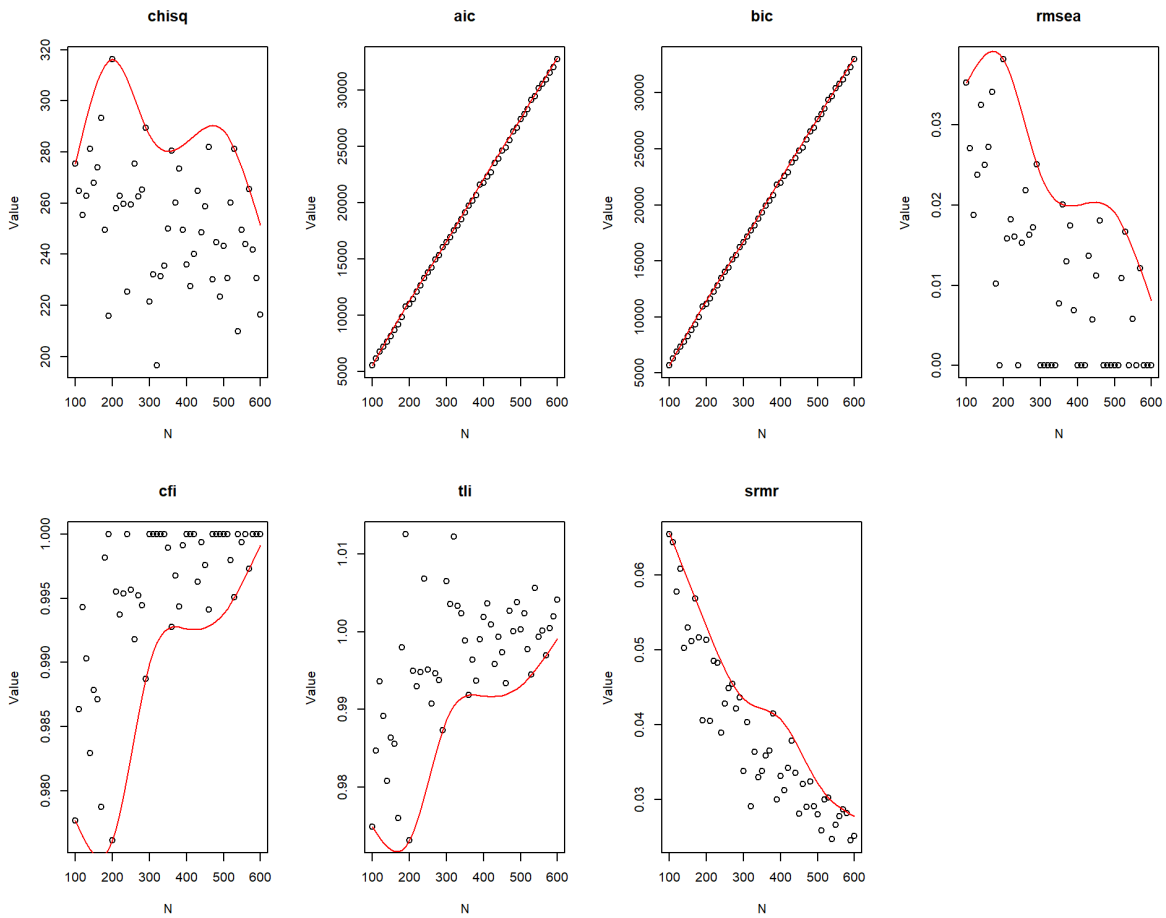


Figure 3: Sampling distributions of fit indices across varying N (ideal conditions)

Source: [Simulation — Ideal Conditions](#)

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Estimate	Power			Std			Average			Rel			Not			Average		
	Av-	(Not	Std	Std	Est	Ave	Average	Average	Rel	Std	SE	Be-	Cove	Not	Average	SD	CI	CI	Width
	er-	Average	Est	Est	Est	SE	Param	Bias	Coverage	Bias	Bias	Bias	low	Above	Width	Width	Width	Width	Width
	age	SD	SE	0)	Est	SD	SE	Param	Bias	Coverage	Bias	Bias	Bias	low	Above	Width	Width	Width	Width
psycho=	0.917	0.034	0.034	1.00	0.920	0.010	0.009	0.920	- 0.92	-	-	0.026	0.02	0.06	0.135	0.004			
	0.003								0.003		0.003	0.095							
psycho=	0.607	0.040	0.037	1.00	0.810	0.019	0.017	0.810	- 0.88	-	-	-	0.02	0.10	0.145	0.004			
	0.003								0.003		0.004	0.073	0.074						
psycho=	0.740	0.031	0.034	1.00	0.940	0.009	0.008	0.940	0.000	0.96	0.000	-	0.093	0.02	0.02	0.133	0.004		
	0.005								0.005										
psycho=	0.732	0.040	0.039	1.00	0.730	0.022	0.022	0.730	0.002	0.96	0.003	0.046	-	0.04	0.00	0.153	0.004		
	0.031								0.031										
psycho=	0.798	0.035	0.039	1.00	0.740	0.018	0.021	0.750	- 0.98	-	-	-	0.094	0.00	0.02	0.151	0.004		
	0.002								0.002		0.003	0.063							
psycho=	0.627	0.049	0.041	1.00	0.630	0.033	0.028	0.630	- 0.88	-	-	-	0.04	0.08	0.159	0.004			
	0.003								0.003		0.004	0.053	0.176						
physical=	0.831	0.036	0.037	1.00	0.840	0.016	0.017	0.840	0.001	0.96	0.001	0.021	0.039	0.02	0.02	0.146	0.004		
physical=	0.829	0.036	0.038	1.00	0.830	0.018	0.017	0.830	- 0.96	-	-	-	0.043	0.02	0.02	0.147	0.004		
	0.001								0.001		0.001	0.034							
physical=	0.623	0.035	0.042	1.00	0.625	0.025	0.029	0.630	- 0.98	-	-	-	0.188	0.00	0.02	0.163	0.004		
	0.007								0.007		0.011	0.206							
physical=	0.713	0.039	0.040	1.00	0.710	0.022	0.024	0.710	0.002	0.98	0.003	0.055	0.026	0.00	0.02	0.157	0.005		
physical=	0.685	0.039	0.041	1.00	0.680	0.029	0.026	0.680	0.005	0.98	0.008	0.137	0.051	0.00	0.02	0.160	0.004		
physical=	0.897	0.041	0.036	1.00	0.890	0.013	0.013	0.890	0.002	0.88	0.002	0.052	-	0.02	0.10	0.141	0.004		
	0.110								0.110										
physical=	0.618	0.044	0.042	1.00	0.610	0.032	0.030	0.610	0.001	0.94	0.002	0.031	-	0.02	0.04	0.164	0.005		
	0.056								0.056										
social=	0.294	0.040	0.035	1.00	0.930	0.012	0.012	0.940	0.002	0.94	0.003	0.059	-	0.02	0.04	0.138	0.004		
	0.122								0.122										
social=	0.289	0.037	0.036	1.00	0.890	0.014	0.014	0.890	0.006	0.96	0.007	0.175	-	0.00	0.04	0.143	0.004		
	0.006								0.006										
social=	0.278	0.040	0.038	1.00	0.782	0.019	0.020	0.780	0.002	0.92	0.003	0.058	-	0.04	0.04	0.151	0.004		
	0.050								0.050										
environment=	0.863	0.049	0.046	1.00	0.362	0.044	0.041	0.370	- 0.94	-	-	-	0.02	0.04	0.179	0.005			
	0.007								0.007		0.020	0.152	0.072						
environment=	0.481	0.040	0.044	1.00	0.483	0.032	0.037	0.490	- 0.98	-	-	-	0.106	0.00	0.02	0.172	0.006		
	0.009								0.009		0.019	0.230							
environment=	0.777	0.020	0.039	1.00	0.782	0.021	0.021	0.790	- 0.94	-	-	-	0.00	0.06	0.152	0.004			
	0.013								0.013		0.016	0.327	0.027						

Table 3: Parameter estimates summary — Ideal conditions, fixed $N = 500$

	Estimate		Power				Not											
	Av- er- age	Estimate SD	Average SE	(Not 0)	Std Est	Std SD	Average SE	Average Parameter	Average Bias	Average Coverage	Rel Bias	Std Bias	SE Bias	Rel Bias	Cove low	Not Above	Average CI Width	SD Width
environmental	0.867	0.031	0.038	1.00	0.812	0.018	0.019	0.810	-	0.94	-	-	-	0.02	0.04	0.14	0.004	
								0.003		0.004	0.08	0.064						
environmental	0.769	0.043	0.039	1.00	0.770	0.022	0.021	0.770	-	0.96	-	-	0.01	0.00	0.04	0.15	0.004	
								0.001		0.002	0.033							
environmental	0.741	0.033	0.040	1.00	0.745	0.019	0.023	0.750	-	0.96	-	-	0.19	0.00	0.04	0.15	0.004	
								0.009		0.012	0.269							
environmental	0.616	0.042	0.042	1.00	0.621	0.028	0.030	0.640	-	0.92	-	-	-	0.00	0.08	0.16	0.005	
								0.024		0.038	0.579	0.005						
environmental	0.728	0.035	0.040	1.00	0.731	0.022	0.024	0.730	-	0.98	-	-	0.12	0.00	0.02	0.15	0.004	
								0.002		0.002	0.049							
psychological	0.292	0.053	0.046	1.00	0.202	0.053	0.046	0.200	0.002	0.90	0.01	0.037	-	0.06	0.04	0.18	0.004	
													0.122					
psychological	0.206	0.041	0.046	1.00	0.206	0.041	0.046	0.200	0.006	0.98	0.02	0.138	0.11	0.00	0.02	0.17	0.003	
psychological	0.204	0.053	0.047	0.98	0.204	0.053	0.047	0.200	0.004	0.94	0.01	0.070	-	0.04	0.02	0.18	0.004	
													0.121					
physical	0.269	0.048	0.046	1.00	0.209	0.048	0.046	0.200	0.009	0.92	0.04	0.179	-	0.06	0.02	0.18	0.004	
													0.028					
physical	0.207	0.053	0.047	0.98	0.207	0.053	0.047	0.200	0.007	0.90	0.03	0.125	-	0.08	0.02	0.18	0.004	
													0.113					
social	0.207	0.040	0.047	1.00	0.207	0.040	0.047	0.200	0.007	0.98	0.03	0.183	0.16	0.02	0.00	0.18	0.003	
Q3~Q4	0.089	0.023	0.020	1.00	0.297	0.058	0.054	0.091	-	0.92	-	-	-	0.04	0.04	0.08	0.004	
								0.002		0.021	0.084	0.103						
Q5~Q5	0.153	0.016	0.015	1.00	0.154	0.018	0.017	0.154	-	0.92	-	-	-	0.04	0.04	0.05	0.004	
								0.001		0.005	0.047	0.106						
Q6~Q6	0.339	0.023	0.024	1.00	0.343	0.031	0.027	0.344	-	0.92	-	-	0.02	0.02	0.06	0.09	0.005	
								0.005		0.015	0.219							
Q7~Q7	0.113	0.016	0.013	1.00	0.114	0.016	0.015	0.116	-	0.92	-	-	-	0.02	0.06	0.05	0.003	
								0.003		0.026	0.193	0.174						
Q11~Q10	0.468	0.031	0.032	1.00	0.466	0.032	0.032	0.467	0.000	0.96	0.00	0.015	0.008	0.00	0.04	0.12	0.008	
Q19~Q19	0.437	0.027	0.030	1.00	0.439	0.027	0.031	0.438	-	0.98	-	-	0.08	0.00	0.02	0.11	0.007	
								0.001		0.001	0.019							
Q26~Q26	0.592	0.032	0.039	1.00	0.601	0.041	0.035	0.603	-	0.94	-	-	0.21	0.00	0.06	0.15	0.008	
								0.011		0.019	0.354							
Q3~Q3	0.285	0.026	0.025	1.00	0.287	0.026	0.028	0.294	-	0.92	-	-	-	0.00	0.08	0.09	0.006	
								0.010		0.033	0.367	0.043						

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Estimate	Power			Std			Average			Rel			Not			Average		
	Av-	Estim	Average	(Not	Std	Est	Ave	Average	Average	Rel	Std	SE	Be-	Cove	Not	Average	SD	CI	SD
	er-	age	SD	SE	0)	Est	SD	SE	Param	Bias	Coverage	Bias	Bias	Bias	low	Above	Width	Width	Width
Q4~Q4	0.308	0.030	0.026	1.00	0.310	0.030	0.029	0.311	- 0.92	-	-	-	0.02	0.06	0.10	0.30	0.006		
									0.003	0.011	0.110	0.119							
Q10~Q10	0.605	0.038	0.041	1.00	0.609	0.031	0.037	0.603	0.002	1.00	0.003	0.052	0.083	0.00	0.00	0.16	0.009		
Q15~Q15	0.493	0.033	0.035	1.00	0.493	0.031	0.035	0.496	- 0.94	-	-	0.033	0.00	0.06	0.13	0.008			
									0.002	0.005	0.072								
Q16~Q16	0.543	0.041	0.037	1.00	0.536	0.039	0.036	0.538	0.005	0.94	0.010	0.130	-	0.00	0.06	0.14	0.010		
												0.080							
Q17~Q17	0.207	0.020	0.021	1.00	0.207	0.024	0.024	0.208	- 0.96	-	-	0.085	0.02	0.02	0.08	0.005			
									0.001	0.003	0.034								
Q18~Q18	0.629	0.046	0.042	1.00	0.627	0.039	0.037	0.628	0.001	0.94	0.001	0.016	-	0.02	0.04	0.16	0.011		
												0.081							
Q20~Q20	0.119	0.022	0.022	1.00	0.118	0.023	0.023	0.116	0.002	1.00	0.018	0.098	0.015	0.00	0.08	0.006			
Q21~Q21	0.208	0.023	0.023	1.00	0.206	0.025	0.025	0.208	0.000	0.94	-	-	0.007	0.04	0.02	0.08	0.005		
												0.001	0.008						
Q22~Q22	0.388	0.028	0.028	1.00	0.389	0.030	0.031	0.392	- 0.92	-	-	0.016	0.00	0.08	0.11	0.007			
									0.003	0.009	0.119								
Q8~Q8	0.869	0.053	0.056	1.00	0.867	0.032	0.030	0.863	0.006	0.98	0.007	0.116	0.063	0.02	0.00	0.21	0.013		
Q9~Q9	0.756	0.052	0.050	1.00	0.765	0.032	0.036	0.760	- 0.94	-	-	-	0.00	0.06	0.19	0.013			
									0.004	0.005	0.071	0.037							
Q12~Q12	0.383	0.031	0.030	1.00	0.388	0.033	0.032	0.376	0.007	0.96	0.018	0.221	-	0.02	0.02	0.11	0.007		
												0.036							
Q13~Q13	0.335	0.026	0.028	1.00	0.340	0.029	0.030	0.344	- 0.94	-	-	0.056	0.00	0.06	0.10	0.007			
									0.009	0.026	0.337								
Q14~Q14	0.405	0.033	0.031	1.00	0.407	0.034	0.033	0.407	- 0.94	-	-	-	0.00	0.06	0.12	0.008			
									0.002	0.004	0.054	0.070							
Q23~Q23	0.440	0.028	0.033	1.00	0.445	0.028	0.034	0.438	0.003	1.00	0.007	0.104	0.164	0.00	0.00	0.12	0.007		
Q24~Q24	0.602	0.037	0.041	1.00	0.614	0.034	0.037	0.590	0.011	1.00	0.019	0.303	0.115	0.00	0.00	0.16	0.009		
Q25~Q25	0.462	0.032	0.034	1.00	0.460	0.032	0.035	0.467	- 0.96	-	-	0.053	0.00	0.04	0.13	0.008			
									0.005	0.011	0.160								

Source: [Simulation — Ideal Conditions](#)

Table 4: Convergence check — Analytical vs. simulation N requirements

Approach	N for 80% Power	Notes
semPower a priori (RMSEA .05)	See Section ??	Model-free, RMSEA-based
semTools a priori (RMSEA .05)	See Section ??	Same underlying math
simsem (parameter power, max N)	From findPower() above	Continuous/ML, population model
simsem (fit power at SRMR < .06)	From getPowerFit() above	Continuous/ML, population model

Source: [Simulation — Ideal Conditions](#)

💡 Convergence Check

If the simulation-based N requirement closely matches the analytical estimate (or is slightly larger to account for sampling error), you have correctly operationalized the model in the simulation structure. This serves as a vital sanity check before introducing complex non-ideal conditions.

5 Monte Carlo Simulation: Realistic Scenarios

Most applied survey data violate the ideal assumptions. Simulating how analytical robustness deteriorates helps us adjust the planned sample sizes to offset the expected penalty.

5.1 Non-Normality

Applying continuous but non-normal (e.g., highly skewed or kurtotic) generation exposes the vulnerability of unadjusted normal-theory ML tests, emphasizing the necessity of robust estimators (e.g., MLR).

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	298.104	5878.811	6042.462	0.038	0.975	0.972	0.065
2	225	292.468	12628.755	12813.149	0.031	0.981	0.979	0.055
3	350	286.833	19378.698	19583.836	0.024	0.987	0.985	0.046

4 475 281.197 26128.641 26354.523 0.017 0.993 0.992 0.036

5 600 275.561 32878.585 33125.209 0.010 0.999 0.999 0.026

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5	0.918	0.049	0.045	
psycho=~Q6	0.819	0.054	0.049	
psycho=~Q7	0.944	0.048	0.044	
psycho=~Q11	0.742	0.057	0.051	
psycho=~Q19	0.749	0.044	0.050	
psycho=~Q26	0.630	0.060	0.053	
physical=~Q3	0.827	0.055	0.048	
physical=~Q4	0.827	0.057	0.049	
physical=~Q10	0.625	0.057	0.054	
physical=~Q15	0.699	0.060	0.053	
physical=~Q16	0.681	0.053	0.052	
physical=~Q17	0.894	0.053	0.047	
physical=~Q18	0.611	0.050	0.054	
social=~Q20	0.930	0.052	0.045	
social=~Q21	0.886	0.050	0.047	
social=~Q22	0.768	0.043	0.050	
environment=~Q8	0.369	0.056	0.059	
environment=~Q9	0.491	0.061	0.057	
environment=~Q12	0.787	0.055	0.050	
environment=~Q13	0.800	0.056	0.050	
environment=~Q14	0.772	0.046	0.051	
environment=~Q23	0.747	0.047	0.052	
environment=~Q24	0.626	0.064	0.054	
environment=~Q25	0.737	0.047	0.052	
psycho~~physical	0.211	0.063	0.060	
psycho~~social	0.209	0.060	0.059	
psycho~~environment	0.217	0.071	0.060	
physical~~social	0.204	0.061	0.060	
physical~~environment	0.194	0.069	0.062	
social~~environment	0.214	0.061	0.061	
Q3~~Q4	0.087	0.026	0.026	
Q5~~Q5	0.154	0.018	0.019	
Q6~~Q6	0.342	0.036	0.032	
Q7~~Q7	0.118	0.020	0.017	
Q11~~Q11	0.470	0.038	0.041	
Q19~~Q19	0.437	0.032	0.039	
Q26~~Q26	0.605	0.039	0.052	
Q3~~Q3	0.281	0.032	0.032	
Q4~~Q4	0.313	0.036	0.035	

Q10~~Q10	0.603	0.043	0.053
Q15~~Q15	0.510	0.052	0.046
Q16~~Q16	0.527	0.054	0.047
Q17~~Q17	0.202	0.025	0.027
Q18~~Q18	0.620	0.057	0.054
Q20~~Q20	0.117	0.032	0.028
Q21~~Q21	0.204	0.030	0.030
Q22~~Q22	0.392	0.032	0.037
Q8~~Q8	0.866	0.073	0.073
Q9~~Q9	0.758	0.070	0.065
Q12~~Q12	0.373	0.040	0.038
Q13~~Q13	0.342	0.038	0.036
Q14~~Q14	0.411	0.042	0.041
Q23~~Q23	0.435	0.039	0.042
Q24~~Q24	0.586	0.044	0.052
Q25~~Q25	0.458	0.042	0.044

	Power (Not equal 0)	Std Est	Std Est SD	Ave SE
psycho==~Q5	1.000	0.919	0.013	0.012
psycho==~Q6	1.000	0.813	0.023	0.021
psycho==~Q7	1.000	0.940	0.010	0.010
psycho==~Q11	1.000	0.733	0.032	0.028
psycho==~Q19	1.000	0.749	0.023	0.027
psycho==~Q26	1.000	0.628	0.042	0.037
physical==~Q3	1.000	0.841	0.023	0.022
physical==~Q4	1.000	0.827	0.026	0.023
physical==~Q10	1.000	0.626	0.040	0.038
physical==~Q15	1.000	0.698	0.039	0.033
physical==~Q16	1.000	0.683	0.040	0.034
physical==~Q17	1.000	0.893	0.016	0.017
physical==~Q18	1.000	0.613	0.039	0.039
social==~Q20	1.000	0.938	0.018	0.016
social==~Q21	1.000	0.891	0.017	0.018
social==~Q22	1.000	0.774	0.025	0.026
environment==~Q8	1.000	0.368	0.053	0.053
environment==~Q9	1.000	0.490	0.054	0.047
environment==~Q12	1.000	0.789	0.028	0.026
environment==~Q13	1.000	0.806	0.024	0.025
environment==~Q14	1.000	0.769	0.025	0.028
environment==~Q23	1.000	0.749	0.022	0.029
environment==~Q24	1.000	0.631	0.039	0.039
environment==~Q25	1.000	0.736	0.029	0.030
psycho~~physical	0.941	0.211	0.063	0.060
psycho~~social	0.961	0.209	0.060	0.059

psycho~~environment	0.922	0.217	0.071	0.060
physical~~social	0.863	0.204	0.061	0.060
physical~~environment	0.824	0.194	0.069	0.062
social~~environment	0.961	0.214	0.061	0.061
Q3~~Q4	0.902	0.292	0.069	0.070
Q5~~Q5	1.000	0.155	0.024	0.022
Q6~~Q6	1.000	0.338	0.038	0.035
Q7~~Q7	1.000	0.117	0.020	0.019
Q11~~Q11	1.000	0.461	0.046	0.042
Q19~~Q19	1.000	0.438	0.034	0.041
Q26~~Q26	1.000	0.604	0.051	0.046
Q3~~Q3	1.000	0.292	0.038	0.036
Q4~~Q4	1.000	0.315	0.043	0.038
Q10~~Q10	1.000	0.607	0.049	0.048
Q15~~Q15	1.000	0.511	0.054	0.045
Q16~~Q16	1.000	0.532	0.055	0.046
Q17~~Q17	1.000	0.203	0.028	0.030
Q18~~Q18	1.000	0.623	0.048	0.048
Q20~~Q20	0.902	0.120	0.034	0.030
Q21~~Q21	1.000	0.206	0.030	0.033
Q22~~Q22	1.000	0.400	0.039	0.041
Q8~~Q8	1.000	0.862	0.039	0.039
Q9~~Q9	1.000	0.757	0.053	0.046
Q12~~Q12	1.000	0.377	0.044	0.041
Q13~~Q13	1.000	0.349	0.040	0.040
Q14~~Q14	1.000	0.408	0.038	0.043
Q23~~Q23	1.000	0.438	0.033	0.044
Q24~~Q24	1.000	0.600	0.049	0.048
Q25~~Q25	1.000	0.458	0.043	0.045
Average Param Average Bias Coverage r_coef.n r_se.n				
psycho==Q5	0.920	-0.002	0.922	-0.234 -0.934
psycho==Q6	0.810	0.009	0.922	-0.337 -0.928
psycho==Q7	0.940	0.004	0.902	-0.227 -0.935
psycho==Q11	0.730	0.012	0.902	-0.414 -0.926
psycho==Q19	0.750	-0.001	0.980	-0.190 -0.931
psycho==Q26	0.630	0.000	0.922	-0.130 -0.936
physical==Q3	0.840	-0.013	0.961	0.172 -0.919
physical==Q4	0.830	-0.003	0.941	0.141 -0.919
physical==Q10	0.630	-0.005	0.922	0.023 -0.935
physical==Q15	0.710	-0.011	0.941	0.091 -0.913
physical==Q16	0.680	0.001	0.961	-0.116 -0.941
physical==Q17	0.890	0.004	0.941	-0.036 -0.913
physical==Q18	0.610	0.001	0.941	-0.167 -0.938

social=~Q20	0.940	-0.010	0.902	-0.071	-0.938
social=~Q21	0.890	-0.004	0.961	-0.075	-0.919
social=~Q22	0.780	-0.012	1.000	0.203	-0.929
environment=~Q8	0.370	-0.001	0.980	0.002	-0.923
environment=~Q9	0.490	0.001	0.980	-0.098	-0.932
environment=~Q12	0.790	-0.003	0.980	-0.190	-0.925
environment=~Q13	0.810	-0.010	0.961	-0.242	-0.925
environment=~Q14	0.770	0.002	1.000	-0.244	-0.923
environment=~Q23	0.750	-0.003	0.941	-0.268	-0.927
environment=~Q24	0.640	-0.014	0.902	-0.020	-0.946
environment=~Q25	0.730	0.007	0.980	-0.179	-0.935
psycho~~physical	0.200	0.011	0.941	-0.059	-0.945
psycho~~social	0.200	0.009	0.941	-0.113	-0.944
psycho~~environment	0.200	0.017	0.902	-0.098	-0.929
physical~~social	0.200	0.004	0.922	-0.126	-0.938
physical~~environment	0.200	-0.006	0.902	-0.050	-0.939
social~~environment	0.200	0.014	0.922	-0.098	-0.932
Q3~~Q4	0.091	-0.003	0.980	-0.107	-0.901
Q5~~Q5	0.154	0.000	0.961	-0.213	-0.895
Q6~~Q6	0.344	-0.002	0.922	0.080	-0.861
Q7~~Q7	0.116	0.001	0.922	-0.226	-0.896
Q11~~Q11	0.467	0.003	0.941	0.055	-0.898
Q19~~Q19	0.438	0.000	0.980	0.006	-0.885
Q26~~Q26	0.603	0.002	0.980	0.068	-0.912
Q3~~Q3	0.294	-0.014	0.941	0.164	-0.879
Q4~~Q4	0.311	0.002	0.980	-0.261	-0.895
Q10~~Q10	0.603	0.000	0.980	0.069	-0.916
Q15~~Q15	0.496	0.015	0.961	-0.114	-0.853
Q16~~Q16	0.538	-0.011	0.922	0.270	-0.908
Q17~~Q17	0.208	-0.006	0.941	0.183	-0.915
Q18~~Q18	0.628	-0.008	0.902	-0.003	-0.903
Q20~~Q20	0.116	0.000	0.941	0.071	-0.909
Q21~~Q21	0.208	-0.004	0.961	-0.166	-0.884
Q22~~Q22	0.392	0.001	0.980	-0.097	-0.895
Q8~~Q8	0.863	0.003	0.980	0.075	-0.874
Q9~~Q9	0.760	-0.002	0.961	0.033	-0.882
Q12~~Q12	0.376	-0.002	0.980	-0.010	-0.875
Q13~~Q13	0.344	-0.002	0.922	0.022	-0.863
Q14~~Q14	0.407	0.004	0.941	0.095	-0.870
Q23~~Q23	0.438	-0.002	0.961	-0.166	-0.894
Q24~~Q24	0.590	-0.005	1.000	-0.035	-0.932
Q25~~Q25	0.467	-0.009	0.922	-0.120	-0.912

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.375	-0.375	0.885	-0.787	-0.895	0.550	-0.373
aic	-0.375	1.000	1.000	-0.649	0.642	0.540	-0.912	1.000
bic	-0.375	1.000	1.000	-0.650	0.643	0.540	-0.912	1.000
rmsea	0.885	-0.649	-0.650	1.000	-0.942	-0.950	0.785	-0.648
cfi	-0.787	0.642	0.643	-0.942	1.000	0.950	-0.774	0.640
tli	-0.895	0.540	0.540	-0.950	0.950	1.000	-0.731	0.537
srmr	0.550	-0.912	-0.912	0.785	-0.774	-0.731	1.000	-0.911
n	-0.373	1.000	1.000	-0.648	0.640	0.537	-0.911	1.000

==== Replications =====

Number of replications = 51

Number of converged replications = 51

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Realistic Scenarios](#)

5.2 Missing Data (MCAR)

Missing data effectively shrinks N through a loss of information, meaning empirical sample sizes must often be adjusted proportionately. This simulation evaluates the explicit power penalty of 10% Missing Completely at Random strings.

RESULT OBJECT

Model Type

[1] "lavaan"

==== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	317.703	5245.664	5479.987	0.045	0.961	0.956	0.068
2	225	307.632	11468.803	11733.948	0.037	0.970	0.967	0.058
3	350	297.562	17691.942	17987.909	0.029	0.980	0.977	0.048
4	475	287.491	23915.081	24241.870	0.021	0.989	0.988	0.038
5	600	277.421	30138.219	30495.831	0.013	0.999	0.999	0.028

==== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5	0.919	0.051	0.046	

psycho=~Q6	0.821	0.056	0.051
psycho=~Q7	0.943	0.050	0.046
psycho=~Q11	0.740	0.059	0.054
psycho=~Q19	0.750	0.047	0.052
psycho=~Q26	0.631	0.062	0.056
physical=~Q3	0.825	0.060	0.049
physical=~Q4	0.825	0.055	0.051
physical=~Q10	0.629	0.063	0.057
physical=~Q15	0.703	0.063	0.055
physical=~Q16	0.676	0.054	0.055
physical=~Q17	0.895	0.056	0.049
physical=~Q18	0.612	0.052	0.057
social=~Q20	0.931	0.056	0.047
social=~Q21	0.888	0.048	0.049
social=~Q22	0.766	0.050	0.052
environment=~Q8	0.370	0.060	0.063
environment=~Q9	0.491	0.058	0.060
environment=~Q12	0.786	0.059	0.053
environment=~Q13	0.801	0.056	0.052
environment=~Q14	0.767	0.049	0.054
environment=~Q23	0.748	0.051	0.054
environment=~Q24	0.625	0.065	0.057
environment=~Q25	0.734	0.054	0.055
psycho~~physical	0.210	0.065	0.060
psycho~~social	0.209	0.063	0.060
psycho~~environment	0.217	0.073	0.061
physical~~social	0.207	0.060	0.061
physical~~environment	0.193	0.068	0.063
social~~environment	0.213	0.065	0.062
Q3~~Q4	0.091	0.027	0.029
Q5~~Q5	0.151	0.021	0.021
Q6~~Q6	0.341	0.033	0.034
Q7~~Q7	0.118	0.022	0.020
Q11~~Q11	0.470	0.041	0.044
Q19~~Q19	0.435	0.036	0.041
Q26~~Q26	0.602	0.046	0.055
Q3~~Q3	0.282	0.034	0.034
Q4~~Q4	0.316	0.038	0.038
Q10~~Q10	0.599	0.047	0.056
Q15~~Q15	0.509	0.048	0.050
Q16~~Q16	0.530	0.048	0.050
Q17~~Q17	0.199	0.027	0.030
Q18~~Q18	0.617	0.057	0.057

Q20~~Q20	0.116	0.035	0.033
Q21~~Q21	0.203	0.035	0.034
Q22~~Q22	0.393	0.032	0.040
Q8~~Q8	0.869	0.073	0.077
Q9~~Q9	0.756	0.069	0.068
Q12~~Q12	0.378	0.043	0.042
Q13~~Q13	0.342	0.038	0.039
Q14~~Q14	0.413	0.048	0.044
Q23~~Q23	0.429	0.038	0.045
Q24~~Q24	0.586	0.049	0.056
Q25~~Q25	0.452	0.044	0.047
Q5~1	-0.009	0.063	0.059
Q6~1	-0.007	0.052	0.060
Q7~1	-0.009	0.062	0.059
Q11~1	-0.004	0.070	0.061
Q19~1	-0.003	0.066	0.060
Q26~1	0.000	0.057	0.060
Q3~1	-0.008	0.063	0.058
Q4~1	-0.012	0.060	0.059
Q10~1	-0.018	0.058	0.060
Q15~1	-0.004	0.065	0.060
Q16~1	-0.020	0.069	0.060
Q17~1	-0.010	0.061	0.059
Q18~1	0.001	0.071	0.060
Q20~1	0.004	0.059	0.059
Q21~1	-0.002	0.054	0.059
Q22~1	0.014	0.072	0.059
Q8~1	-0.012	0.061	0.061
Q9~1	-0.004	0.064	0.061
Q12~1	-0.005	0.075	0.060
Q13~1	-0.018	0.064	0.059
Q14~1	-0.009	0.057	0.060
Q23~1	0.005	0.060	0.060
Q24~1	-0.014	0.065	0.060
Q25~1	-0.005	0.070	0.060

	Power (Not equal 0)	Std Est	Std Est SD	Std Ave SE
psycho=~Q5	1.000	0.921	0.014	0.013
psycho=~Q6	1.000	0.814	0.023	0.023
psycho=~Q7	1.000	0.939	0.011	0.011
psycho=~Q11	1.000	0.732	0.034	0.030
psycho=~Q19	1.000	0.750	0.026	0.029
psycho=~Q26	1.000	0.629	0.044	0.039
physical=~Q3	1.000	0.839	0.026	0.023

physical=~Q4	1.000	0.825	0.025	0.025
physical=~Q10	1.000	0.629	0.044	0.040
physical=~Q15	1.000	0.700	0.039	0.035
physical=~Q16	1.000	0.679	0.037	0.036
physical=~Q17	1.000	0.894	0.018	0.018
physical=~Q18	1.000	0.614	0.039	0.041
social=~Q20	1.000	0.938	0.020	0.019
social=~Q21	1.000	0.892	0.019	0.020
social=~Q22	1.000	0.772	0.029	0.028
environment=~Q8	1.000	0.369	0.057	0.057
environment=~Q9	1.000	0.491	0.052	0.050
environment=~Q12	1.000	0.787	0.029	0.028
environment=~Q13	1.000	0.807	0.025	0.027
environment=~Q14	1.000	0.766	0.027	0.030
environment=~Q23	1.000	0.752	0.027	0.031
environment=~Q24	1.000	0.631	0.040	0.041
environment=~Q25	1.000	0.736	0.033	0.033
psycho~~physical	0.902	0.210	0.065	0.060
psycho~~social	0.961	0.209	0.063	0.060
psycho~~environment	0.922	0.217	0.073	0.061
physical~~social	0.882	0.207	0.060	0.061
physical~~environment	0.843	0.193	0.068	0.063
social~~environment	0.961	0.213	0.065	0.062
Q3~~Q4	0.922	0.301	0.072	0.077
Q5~~Q5	1.000	0.152	0.025	0.023
Q6~~Q6	1.000	0.337	0.037	0.037
Q7~~Q7	1.000	0.118	0.021	0.021
Q11~~Q11	1.000	0.463	0.049	0.044
Q19~~Q19	1.000	0.437	0.039	0.043
Q26~~Q26	1.000	0.602	0.054	0.048
Q3~~Q3	1.000	0.295	0.043	0.039
Q4~~Q4	1.000	0.318	0.041	0.041
Q10~~Q10	1.000	0.603	0.055	0.050
Q15~~Q15	1.000	0.508	0.054	0.048
Q16~~Q16	1.000	0.537	0.050	0.049
Q17~~Q17	1.000	0.200	0.032	0.033
Q18~~Q18	1.000	0.622	0.049	0.051
Q20~~Q20	0.941	0.119	0.038	0.035
Q21~~Q21	1.000	0.205	0.033	0.036
Q22~~Q22	1.000	0.403	0.045	0.044
Q8~~Q8	1.000	0.861	0.043	0.042
Q9~~Q9	1.000	0.756	0.052	0.049
Q12~~Q12	1.000	0.380	0.045	0.044

Q13~~Q13	1.000	0.348	0.040	0.043
Q14~~Q14	1.000	0.413	0.042	0.046
Q23~~Q23	1.000	0.434	0.040	0.047
Q24~~Q24	1.000	0.601	0.050	0.051
Q25~~Q25	1.000	0.457	0.049	0.048
Q5~1	0.078	-0.008	0.062	0.059
Q6~1	0.039	-0.007	0.052	0.059
Q7~1	0.078	-0.008	0.061	0.059
Q11~1	0.039	-0.003	0.069	0.060
Q19~1	0.020	-0.002	0.066	0.060
Q26~1	0.039	0.000	0.057	0.060
Q3~1	0.078	-0.007	0.064	0.059
Q4~1	0.098	-0.012	0.060	0.059
Q10~1	0.039	-0.017	0.058	0.060
Q15~1	0.059	-0.003	0.065	0.060
Q16~1	0.118	-0.020	0.070	0.060
Q17~1	0.078	-0.011	0.060	0.059
Q18~1	0.098	0.001	0.071	0.060
Q20~1	0.059	0.004	0.060	0.059
Q21~1	0.039	-0.002	0.055	0.059
Q22~1	0.098	0.014	0.073	0.060
Q8~1	0.039	-0.011	0.061	0.061
Q9~1	0.020	-0.004	0.064	0.061
Q12~1	0.059	-0.006	0.074	0.060
Q13~1	0.059	-0.020	0.068	0.060
Q14~1	0.039	-0.010	0.058	0.060
Q23~1	0.000	0.005	0.060	0.060
Q24~1	0.098	-0.014	0.068	0.060
Q25~1	0.078	-0.005	0.070	0.060
Average FMI1 SD FMI1 r_coef.n r_se.n				
psycho=~Q5	0.046	0.013	-0.224	-0.925
psycho=~Q6	0.067	0.019	-0.386	-0.925
psycho=~Q7	0.047	0.015	-0.163	-0.935
psycho=~Q11	0.079	0.026	-0.351	-0.922
psycho=~Q19	0.074	0.021	-0.145	-0.931
psycho=~Q26	0.087	0.023	-0.161	-0.922
physical=~Q3	0.066	0.020	0.191	-0.924
physical=~Q4	0.068	0.025	0.131	-0.914
physical=~Q10	0.094	0.021	-0.006	-0.940
physical=~Q15	0.087	0.020	0.046	-0.908
physical=~Q16	0.084	0.029	-0.080	-0.929
physical=~Q17	0.073	0.022	-0.062	-0.915
physical=~Q18	0.094	0.035	-0.271	-0.942

social=~Q20	0.077	0.022	-0.076	-0.934
social=~Q21	0.072	0.023	-0.065	-0.919
social=~Q22	0.078	0.023	0.216	-0.925
environment=~Q8	0.104	0.032	-0.038	-0.913
environment=~Q9	0.098	0.028	-0.137	-0.931
environment=~Q12	0.079	0.026	-0.120	-0.910
environment=~Q13	0.079	0.019	-0.205	-0.931
environment=~Q14	0.090	0.028	-0.210	-0.916
environment=~Q23	0.083	0.028	-0.297	-0.931
environment=~Q24	0.092	0.025	0.022	-0.939
environment=~Q25	0.094	0.028	-0.132	-0.924
psycho~~physical	0.021	0.008	-0.035	-0.945
psycho~~social	0.025	0.008	-0.099	-0.940
psycho~~environment	0.023	0.008	-0.115	-0.928
physical~~social	0.030	0.011	-0.129	-0.938
physical~~environment	0.024	0.011	-0.038	-0.938
social~~environment	0.031	0.011	-0.073	-0.933
Q3~~Q4	0.190	0.034	-0.244	-0.912
Q5~~Q5	0.178	0.024	-0.103	-0.875
Q6~~Q6	0.139	0.023	0.125	-0.880
Q7~~Q7	0.193	0.033	-0.282	-0.891
Q11~~Q11	0.121	0.021	-0.027	-0.884
Q19~~Q19	0.125	0.022	-0.008	-0.889
Q26~~Q26	0.113	0.025	0.197	-0.877
Q3~~Q3	0.157	0.031	0.096	-0.886
Q4~~Q4	0.154	0.025	-0.329	-0.893
Q10~~Q10	0.121	0.024	0.166	-0.916
Q15~~Q15	0.128	0.024	-0.124	-0.867
Q16~~Q16	0.118	0.027	0.257	-0.905
Q17~~Q17	0.171	0.023	0.184	-0.897
Q18~~Q18	0.118	0.020	0.077	-0.912
Q20~~Q20	0.230	0.038	0.059	-0.894
Q21~~Q21	0.211	0.038	-0.110	-0.882
Q22~~Q22	0.147	0.039	-0.135	-0.899
Q8~~Q8	0.111	0.031	0.069	-0.881
Q9~~Q9	0.117	0.025	0.097	-0.893
Q12~~Q12	0.149	0.032	-0.137	-0.853
Q13~~Q13	0.157	0.037	0.072	-0.880
Q14~~Q14	0.146	0.029	0.041	-0.855
Q23~~Q23	0.140	0.022	0.016	-0.892
Q24~~Q24	0.121	0.019	-0.066	-0.924
Q25~~Q25	0.143	0.039	-0.049	-0.908
Q5~1	0.024	0.005	0.170	-0.928

Q6~1	0.040	0.008	0.196	-0.928
Q7~1	0.022	0.005	0.135	-0.932
Q11~1	0.051	0.011	0.086	-0.923
Q19~1	0.051	0.010	0.099	-0.937
Q26~1	0.065	0.010	0.192	-0.924
Q3~1	0.032	0.008	-0.024	-0.908
Q4~1	0.034	0.008	-0.043	-0.907
Q10~1	0.068	0.011	0.301	-0.929
Q15~1	0.061	0.012	0.058	-0.900
Q16~1	0.059	0.014	0.181	-0.920
Q17~1	0.033	0.009	0.060	-0.902
Q18~1	0.069	0.013	0.083	-0.935
Q20~1	0.030	0.007	-0.222	-0.931
Q21~1	0.034	0.007	-0.034	-0.922
Q22~1	0.046	0.010	-0.099	-0.925
Q8~1	0.090	0.014	0.085	-0.922
Q9~1	0.083	0.018	-0.139	-0.939
Q12~1	0.050	0.012	-0.219	-0.916
Q13~1	0.047	0.011	0.034	-0.935
Q14~1	0.055	0.011	-0.254	-0.925
Q23~1	0.053	0.011	-0.271	-0.924
Q24~1	0.067	0.013	-0.265	-0.937
Q25~1	0.057	0.012	-0.117	-0.926

==== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.309	-0.310	0.929	-0.787	-0.860	0.509	-0.307
aic	-0.309	1.000	1.000	-0.470	0.466	0.437	-0.910	1.000
bic	-0.310	1.000	1.000	-0.470	0.466	0.437	-0.910	1.000
rmsea	0.929	-0.470	-0.470	1.000	-0.906	-0.938	0.637	-0.468
cfi	-0.787	0.466	0.466	-0.906	1.000	0.984	-0.644	0.464
tli	-0.860	0.437	0.437	-0.938	0.984	1.000	-0.641	0.435
srmr	0.509	-0.910	-0.910	0.637	-0.644	-0.641	1.000	-0.908
n	-0.307	1.000	1.000	-0.468	0.464	0.435	-0.908	1.000

==== Replications =====

Number of replications = 51

Number of converged replications = 50

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 1

NOTE: The sample size is varying.

NOTE: The data generation model is not the same as the analysis model. See the summary of t

Source: [Simulation — Realistic Scenarios](#)

5.3 Sensitivity Across Population Models

A robust power analysis examines boundaries. How drastically does necessary N shift when our assumption about structural quality is overly naive, or overly optimistic?

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	305.230	6919.921	7083.572	0.038	0.896	0.883	0.074
2	225	297.615	14983.600	15167.995	0.031	0.921	0.911	0.064
3	350	290.001	23047.280	23252.418	0.025	0.945	0.938	0.053
4	475	282.386	31110.959	31336.840	0.019	0.969	0.965	0.042
5	600	274.772	39174.638	39421.263	0.013	0.994	0.993	0.032

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5	0.545		0.068	0.064
psycho=~Q6	0.522		0.055	0.064
psycho=~Q7	0.549		0.056	0.065
psycho=~Q11	0.522		0.055	0.064
psycho=~Q19	0.514		0.064	0.065
psycho=~Q26	0.519		0.064	0.065
physical=~Q3	0.486		0.078	0.068
physical=~Q4	0.467		0.080	0.069
physical=~Q10	0.477		0.073	0.067
physical=~Q15	0.471		0.068	0.067
physical=~Q16	0.485		0.080	0.067
physical=~Q17	0.474		0.092	0.067
physical=~Q18	0.493		0.060	0.067
social=~Q20	0.596		0.077	0.071
social=~Q21	0.590		0.078	0.070
social=~Q22	0.598		0.077	0.071
environment=~Q8	0.474		0.082	0.065
environment=~Q9	0.489		0.078	0.064
environment=~Q12	0.481		0.052	0.065
environment=~Q13	0.484		0.073	0.064

environment=~Q14	0.478	0.064	0.064
environment=~Q23	0.493	0.049	0.064
environment=~Q24	0.474	0.066	0.064
environment=~Q25	0.482	0.054	0.065
psycho~~physical	0.317	0.079	0.079
psycho~~social	0.302	0.092	0.082
psycho~~environment	0.295	0.087	0.077
physical~~social	0.321	0.092	0.083
physical~~environment	0.315	0.081	0.078
social~~environment	0.312	0.099	0.081
Q3~~Q4	0.236	0.058	0.058
Q5~~Q5	0.694	0.082	0.069
Q6~~Q6	0.711	0.074	0.069
Q7~~Q7	0.708	0.071	0.071
Q11~~Q11	0.715	0.065	0.069
Q19~~Q19	0.729	0.072	0.071
Q26~~Q26	0.731	0.064	0.071
Q3~~Q3	0.772	0.067	0.075
Q4~~Q4	0.787	0.063	0.076
Q10~~Q10	0.764	0.069	0.073
Q15~~Q15	0.773	0.072	0.073
Q16~~Q16	0.756	0.072	0.073
Q17~~Q17	0.774	0.072	0.074
Q18~~Q18	0.768	0.075	0.074
Q20~~Q20	0.641	0.086	0.078
Q21~~Q21	0.611	0.096	0.076
Q22~~Q22	0.632	0.102	0.078
Q8~~Q8	0.774	0.086	0.072
Q9~~Q9	0.758	0.071	0.071
Q12~~Q12	0.768	0.067	0.071
Q13~~Q13	0.758	0.075	0.071
Q14~~Q14	0.765	0.067	0.071
Q23~~Q23	0.744	0.063	0.070
Q24~~Q24	0.759	0.073	0.071
Q25~~Q25	0.765	0.070	0.071

	Power (Not equal 0)	Std Est	Std Est SD	Std Ave SE
psycho=~Q5	1.000	0.547	0.056	0.054
psycho=~Q6	1.000	0.526	0.052	0.055
psycho=~Q7	1.000	0.546	0.045	0.054
psycho=~Q11	1.000	0.524	0.043	0.055
psycho=~Q19	1.000	0.515	0.054	0.055
psycho=~Q26	1.000	0.517	0.054	0.055
physical=~Q3	1.000	0.482	0.068	0.060

physical=~Q4	1.000	0.463	0.068	0.061
physical=~Q10	1.000	0.477	0.062	0.059
physical=~Q15	1.000	0.471	0.059	0.059
physical=~Q16	1.000	0.485	0.067	0.059
physical=~Q17	0.980	0.471	0.077	0.059
physical=~Q18	1.000	0.489	0.053	0.058
social=~Q20	1.000	0.595	0.068	0.060
social=~Q21	1.000	0.601	0.073	0.061
social=~Q22	1.000	0.600	0.073	0.060
environment=~Q8	1.000	0.472	0.070	0.057
environment=~Q9	1.000	0.488	0.068	0.056
environment=~Q12	1.000	0.480	0.048	0.056
environment=~Q13	1.000	0.484	0.061	0.056
environment=~Q14	1.000	0.478	0.049	0.056
environment=~Q23	1.000	0.496	0.047	0.055
environment=~Q24	1.000	0.477	0.058	0.057
environment=~Q25	1.000	0.482	0.049	0.056
psycho~~physical	0.961	0.317	0.079	0.079
psycho~~social	0.922	0.302	0.092	0.082
psycho~~environment	0.941	0.295	0.087	0.077
physical~~social	0.941	0.321	0.092	0.083
physical~~environment	1.000	0.315	0.081	0.078
social~~environment	0.863	0.312	0.099	0.081
Q3~~Q4	0.980	0.301	0.065	0.061
Q5~~Q5	1.000	0.698	0.061	0.059
Q6~~Q6	1.000	0.720	0.055	0.057
Q7~~Q7	1.000	0.700	0.050	0.059
Q11~~Q11	1.000	0.724	0.044	0.058
Q19~~Q19	1.000	0.732	0.056	0.056
Q26~~Q26	1.000	0.730	0.053	0.056
Q3~~Q3	1.000	0.763	0.064	0.058
Q4~~Q4	1.000	0.781	0.060	0.056
Q10~~Q10	1.000	0.769	0.061	0.056
Q15~~Q15	1.000	0.775	0.055	0.056
Q16~~Q16	1.000	0.760	0.066	0.057
Q17~~Q17	1.000	0.772	0.073	0.055
Q18~~Q18	1.000	0.758	0.051	0.057
Q20~~Q20	1.000	0.641	0.080	0.071
Q21~~Q21	0.980	0.633	0.089	0.074
Q22~~Q22	1.000	0.635	0.087	0.072
Q8~~Q8	1.000	0.772	0.067	0.053
Q9~~Q9	1.000	0.757	0.067	0.054
Q12~~Q12	1.000	0.767	0.045	0.054

Q13~~Q13	1.000	0.762	0.063	0.054
Q14~~Q14	1.000	0.769	0.048	0.054
Q23~~Q23	1.000	0.752	0.045	0.055
Q24~~Q24	1.000	0.769	0.052	0.053
Q25~~Q25	1.000	0.765	0.048	0.054
	Average Param	Average Bias	Coverage	r_coef.n r_se.n
psycho=~Q5	0.533	0.013	0.941	-0.448 -0.888
psycho=~Q6	0.533	-0.010	0.961	-0.030 -0.928
psycho=~Q7	0.533	0.017	0.980	-0.235 -0.908
psycho=~Q11	0.533	-0.011	0.961	-0.219 -0.941
psycho=~Q19	0.533	-0.019	0.941	0.171 -0.917
psycho=~Q26	0.533	-0.014	0.980	0.083 -0.936
physical=~Q3	0.476	0.010	0.961	-0.256 -0.933
physical=~Q4	0.476	-0.009	0.961	-0.111 -0.940
physical=~Q10	0.476	0.001	0.941	-0.065 -0.918
physical=~Q15	0.476	-0.005	0.961	-0.116 -0.937
physical=~Q16	0.476	0.009	0.941	-0.067 -0.925
physical=~Q17	0.476	-0.002	0.882	-0.177 -0.931
physical=~Q18	0.476	0.017	0.980	-0.311 -0.937
social=~Q20	0.602	-0.006	0.941	-0.072 -0.897
social=~Q21	0.602	-0.011	0.922	0.017 -0.882
social=~Q22	0.602	-0.004	0.941	0.003 -0.905
environment=~Q8	0.483	-0.009	0.941	0.034 -0.919
environment=~Q9	0.483	0.006	0.961	-0.051 -0.948
environment=~Q12	0.483	-0.003	0.961	0.153 -0.943
environment=~Q13	0.483	0.001	0.922	-0.147 -0.936
environment=~Q14	0.483	-0.005	0.961	-0.019 -0.922
environment=~Q23	0.483	0.010	0.961	-0.287 -0.937
environment=~Q24	0.483	-0.009	0.980	0.234 -0.923
environment=~Q25	0.483	-0.001	0.941	0.076 -0.931
psycho~~physical	0.300	0.017	0.941	0.043 -0.923
psycho~~social	0.300	0.002	0.902	-0.084 -0.923
psycho~~environment	0.300	-0.005	0.941	0.023 -0.919
physical~~social	0.300	0.021	0.941	-0.210 -0.929
physical~~environment	0.300	0.015	0.902	-0.162 -0.938
social~~environment	0.300	0.012	0.882	-0.004 -0.933
Q3~~Q4	0.232	0.003	0.980	0.141 -0.925
Q5~~Q5	0.716	-0.022	0.902	0.080 -0.816
Q6~~Q6	0.716	-0.006	0.961	0.195 -0.890
Q7~~Q7	0.716	-0.008	0.922	0.019 -0.859
Q11~~Q11	0.716	-0.001	1.000	-0.014 -0.918
Q19~~Q19	0.716	0.013	0.980	-0.330 -0.874
Q26~~Q26	0.716	0.015	0.961	-0.351 -0.914

Q3~~Q3	0.774	-0.002	0.961	0.127	-0.903
Q4~~Q4	0.774	0.014	0.980	-0.357	-0.925
Q10~~Q10	0.774	-0.010	1.000	-0.007	-0.876
Q15~~Q15	0.774	-0.001	0.961	0.071	-0.906
Q16~~Q16	0.774	-0.018	0.941	0.191	-0.881
Q17~~Q17	0.774	0.001	0.961	0.072	-0.889
Q18~~Q18	0.774	-0.006	0.961	0.178	-0.906
Q20~~Q20	0.638	0.003	0.941	-0.062	-0.840
Q21~~Q21	0.638	-0.027	0.961	0.151	-0.792
Q22~~Q22	0.638	-0.006	0.922	-0.059	-0.855
Q8~~Q8	0.766	0.008	0.902	0.071	-0.861
Q9~~Q9	0.766	-0.009	0.961	0.035	-0.922
Q12~~Q12	0.766	0.001	0.941	-0.064	-0.920
Q13~~Q13	0.766	-0.008	0.941	0.273	-0.893
Q14~~Q14	0.766	-0.002	0.980	0.119	-0.872
Q23~~Q23	0.766	-0.023	0.980	0.058	-0.914
Q24~~Q24	0.766	-0.008	0.941	-0.242	-0.877
Q25~~Q25	0.766	-0.002	0.922	-0.111	-0.892

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.351	-0.351	0.886	-0.785	-0.902	0.517	-0.350
aic	-0.351	1.000	1.000	-0.646	0.652	0.508	-0.926	1.000
bic	-0.351	1.000	1.000	-0.647	0.652	0.508	-0.926	1.000
rmsea	0.886	-0.646	-0.647	1.000	-0.946	-0.943	0.791	-0.645
cfi	-0.785	0.652	0.652	-0.946	1.000	0.934	-0.800	0.650
tli	-0.902	0.508	0.508	-0.943	0.934	1.000	-0.708	0.505
srmr	0.517	-0.926	-0.926	0.791	-0.800	-0.708	1.000	-0.926
n	-0.350	1.000	1.000	-0.645	0.650	0.505	-0.926	1.000

===== Replications =====

Number of replications = 51

Number of converged replications = 51

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Realistic Scenarios](#)

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	326.336	6401.706	6565.357	0.041	0.953	0.947	0.069
2	225	312.106	13816.069	14000.463	0.034	0.965	0.960	0.059
3	350	297.875	21230.432	21435.570	0.027	0.976	0.973	0.049
4	475	283.644	28644.795	28870.676	0.020	0.988	0.987	0.038
5	600	269.413	36059.157	36305.782	0.012	1.000	1.000	0.028

===== Parameter Estimates and Standard Errors =====

	Estimate	Average	Estimate	SD	Average	SE
psycho=~Q5	0.693		0.047		0.054	
psycho=~Q6	0.703		0.044		0.054	
psycho=~Q7	0.712		0.055		0.054	
psycho=~Q11	0.690		0.055		0.053	
psycho=~Q19	0.707		0.048		0.054	
psycho=~Q26	0.711		0.053		0.053	
physical=~Q3	0.697		0.062		0.055	
physical=~Q4	0.682		0.058		0.056	
physical=~Q10	0.697		0.056		0.055	
physical=~Q15	0.682		0.061		0.054	
physical=~Q16	0.680		0.055		0.055	
physical=~Q17	0.689		0.051		0.055	
physical=~Q18	0.691		0.055		0.055	
social=~Q20	0.762		0.046		0.056	
social=~Q21	0.742		0.064		0.055	
social=~Q22	0.763		0.063		0.056	
environment=~Q8	0.648		0.053		0.056	
environment=~Q9	0.647		0.062		0.056	
environment=~Q12	0.641		0.051		0.056	
environment=~Q13	0.637		0.059		0.056	
environment=~Q14	0.641		0.047		0.056	
environment=~Q23	0.635		0.064		0.056	
environment=~Q24	0.645		0.063		0.055	
environment=~Q25	0.644		0.067		0.056	
psycho~~physical	0.086		0.069		0.067	
psycho~~social	0.075		0.084		0.069	
psycho~~environment	0.075		0.069		0.067	
physical~~social	0.085		0.064		0.069	
physical~~environment	0.091		0.076		0.067	
social~~environment	0.086		0.075		0.069	
Q3~~Q4	0.161		0.045		0.041	
Q5~~Q5	0.499		0.048		0.049	

Q6~~Q6	0.501	0.041	0.050
Q7~~Q7	0.493	0.049	0.049
Q11~~Q11	0.494	0.051	0.049
Q19~~Q19	0.498	0.060	0.050
Q26~~Q26	0.476	0.040	0.048
Q3~~Q3	0.532	0.048	0.053
Q4~~Q4	0.543	0.046	0.054
Q10~~Q10	0.523	0.041	0.051
Q15~~Q15	0.522	0.049	0.050
Q16~~Q16	0.530	0.059	0.051
Q17~~Q17	0.525	0.051	0.051
Q18~~Q18	0.534	0.047	0.052
Q20~~Q20	0.410	0.065	0.054
Q21~~Q21	0.403	0.059	0.052
Q22~~Q22	0.410	0.056	0.054
Q8~~Q8	0.586	0.058	0.055
Q9~~Q9	0.588	0.067	0.055
Q12~~Q12	0.591	0.056	0.055
Q13~~Q13	0.586	0.060	0.055
Q14~~Q14	0.592	0.069	0.056
Q23~~Q23	0.588	0.059	0.055
Q24~~Q24	0.570	0.061	0.054
Q25~~Q25	0.582	0.057	0.055

	Power (Not equal 0)	Std Est	Std Est SD	Ave SE
psycho=~Q5	1.000	0.700	0.029	0.035
psycho=~Q6	1.000	0.704	0.026	0.035
psycho=~Q7	1.000	0.711	0.035	0.034
psycho=~Q11	1.000	0.700	0.038	0.036
psycho=~Q19	1.000	0.707	0.034	0.035
psycho=~Q26	1.000	0.717	0.030	0.034
physical=~Q3	1.000	0.689	0.039	0.037
physical=~Q4	1.000	0.678	0.038	0.038
physical=~Q10	1.000	0.693	0.036	0.036
physical=~Q15	1.000	0.685	0.037	0.036
physical=~Q16	1.000	0.682	0.037	0.037
physical=~Q17	1.000	0.689	0.034	0.036
physical=~Q18	1.000	0.686	0.036	0.036
social=~Q20	1.000	0.765	0.033	0.037
social=~Q21	1.000	0.758	0.045	0.037
social=~Q22	1.000	0.765	0.040	0.037
environment=~Q8	1.000	0.645	0.041	0.039
environment=~Q9	1.000	0.644	0.048	0.040
environment=~Q12	1.000	0.640	0.036	0.040

environment=~Q13	1.000	0.638	0.038	0.040
environment=~Q14	1.000	0.640	0.036	0.040
environment=~Q23	1.000	0.636	0.045	0.040
environment=~Q24	1.000	0.648	0.042	0.039
environment=~Q25	1.000	0.643	0.050	0.040
psycho~~physical	0.333	0.086	0.069	0.067
psycho~~social	0.235	0.075	0.084	0.069
psycho~~environment	0.235	0.075	0.069	0.067
physical~~social	0.196	0.085	0.064	0.069
physical~~environment	0.294	0.091	0.076	0.067
social~~environment	0.275	0.086	0.075	0.069
Q3~~Q4	0.941	0.298	0.070	0.062
Q5~~Q5	1.000	0.510	0.041	0.049
Q6~~Q6	1.000	0.504	0.036	0.049
Q7~~Q7	1.000	0.493	0.050	0.049
Q11~~Q11	1.000	0.509	0.051	0.049
Q19~~Q19	1.000	0.499	0.048	0.049
Q26~~Q26	1.000	0.485	0.043	0.049
Q3~~Q3	1.000	0.524	0.054	0.051
Q4~~Q4	1.000	0.539	0.051	0.051
Q10~~Q10	1.000	0.519	0.050	0.050
Q15~~Q15	1.000	0.529	0.051	0.050
Q16~~Q16	1.000	0.534	0.050	0.050
Q17~~Q17	1.000	0.525	0.046	0.050
Q18~~Q18	1.000	0.528	0.049	0.050
Q20~~Q20	1.000	0.413	0.051	0.056
Q21~~Q21	1.000	0.423	0.067	0.056
Q22~~Q22	1.000	0.414	0.061	0.056
Q8~~Q8	1.000	0.583	0.054	0.051
Q9~~Q9	1.000	0.583	0.061	0.051
Q12~~Q12	1.000	0.590	0.045	0.051
Q13~~Q13	1.000	0.591	0.049	0.051
Q14~~Q14	1.000	0.589	0.045	0.051
Q23~~Q23	1.000	0.593	0.059	0.051
Q24~~Q24	1.000	0.578	0.056	0.051
Q25~~Q25	1.000	0.584	0.063	0.051
Average Param Average Bias Coverage r_coef.n r_se.n				
psycho=~Q5	0.711	-0.019	0.902	-0.171 -0.942
psycho=~Q6	0.711	-0.008	1.000	-0.121 -0.932
psycho=~Q7	0.711	0.001	0.941	-0.313 -0.927
psycho=~Q11	0.711	-0.021	0.941	0.135 -0.923
psycho=~Q19	0.711	-0.004	0.961	-0.052 -0.907
psycho=~Q26	0.711	0.000	0.941	-0.326 -0.921

physical=~Q3	0.684	0.013	0.961	-0.353	-0.935
physical=~Q4	0.684	-0.002	0.961	-0.224	-0.941
physical=~Q10	0.684	0.014	0.980	-0.092	-0.944
physical=~Q15	0.684	-0.001	0.902	-0.100	-0.940
physical=~Q16	0.684	-0.004	0.961	-0.081	-0.929
physical=~Q17	0.684	0.006	0.961	-0.441	-0.926
physical=~Q18	0.684	0.008	0.961	-0.221	-0.920
social=~Q20	0.766	-0.004	1.000	-0.022	-0.911
social=~Q21	0.766	-0.024	0.882	0.120	-0.937
social=~Q22	0.766	-0.003	0.922	-0.037	-0.914
environment=~Q8	0.642	0.005	0.922	-0.217	-0.923
environment=~Q9	0.642	0.004	0.941	-0.162	-0.938
environment=~Q12	0.642	-0.002	0.980	0.209	-0.931
environment=~Q13	0.642	-0.006	0.902	0.146	-0.913
environment=~Q14	0.642	-0.002	1.000	0.037	-0.918
environment=~Q23	0.642	-0.008	0.922	-0.090	-0.933
environment=~Q24	0.642	0.003	0.941	0.000	-0.925
environment=~Q25	0.642	0.001	0.882	-0.005	-0.942
psycho~~physical	0.080	0.006	0.980	0.017	-0.943
psycho~~social	0.080	-0.005	0.941	-0.018	-0.943
psycho~~environment	0.080	-0.005	0.961	0.079	-0.946
physical~~social	0.080	0.005	0.980	-0.180	-0.938
physical~~environment	0.080	0.011	0.882	-0.191	-0.943
social~~environment	0.080	0.006	0.941	0.020	-0.942
Q3~~Q4	0.160	0.002	0.961	0.155	-0.919
Q5~~Q5	0.494	0.004	0.961	-0.095	-0.916
Q6~~Q6	0.494	0.007	0.980	-0.136	-0.906
Q7~~Q7	0.494	-0.001	0.980	-0.154	-0.888
Q11~~Q11	0.494	0.000	0.941	-0.008	-0.865
Q19~~Q19	0.494	0.004	0.902	-0.084	-0.827
Q26~~Q26	0.494	-0.018	1.000	0.130	-0.881
Q3~~Q3	0.533	-0.001	1.000	0.133	-0.901
Q4~~Q4	0.533	0.010	0.961	-0.310	-0.926
Q10~~Q10	0.533	-0.010	0.941	0.114	-0.915
Q15~~Q15	0.533	-0.011	1.000	0.198	-0.909
Q16~~Q16	0.533	-0.002	0.902	-0.009	-0.882
Q17~~Q17	0.533	-0.008	0.922	0.017	-0.888
Q18~~Q18	0.533	0.002	0.961	-0.159	-0.878
Q20~~Q20	0.413	-0.003	0.902	-0.069	-0.857
Q21~~Q21	0.413	-0.010	0.941	0.119	-0.898
Q22~~Q22	0.413	-0.003	0.922	-0.060	-0.856
Q8~~Q8	0.587	-0.001	0.941	0.228	-0.877
Q9~~Q9	0.587	0.000	0.980	-0.029	-0.886

Q12~~Q12	0.587	0.004	0.980	0.159	-0.874
Q13~~Q13	0.587	-0.001	0.922	-0.173	-0.861
Q14~~Q14	0.587	0.004	0.902	-0.177	-0.858
Q23~~Q23	0.587	0.000	0.941	0.137	-0.894
Q24~~Q24	0.587	-0.017	0.922	0.079	-0.865
Q25~~Q25	0.587	-0.006	0.941	0.140	-0.903

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.303	-0.303	0.889	-0.785	-0.911	0.485	-0.301
aic	-0.303	1.000	1.000	-0.620	0.597	0.454	-0.919	1.000
bic	-0.303	1.000	1.000	-0.620	0.597	0.454	-0.919	1.000
rmsea	0.889	-0.620	-0.620	1.000	-0.937	-0.936	0.752	-0.618
cfi	-0.785	0.597	0.597	-0.937	1.000	0.925	-0.718	0.595
tli	-0.911	0.454	0.454	-0.936	0.925	1.000	-0.656	0.451
srmr	0.485	-0.919	-0.919	0.752	-0.718	-0.656	1.000	-0.918
n	-0.301	1.000	1.000	-0.618	0.595	0.451	-0.918	1.000

===== Replications =====

Number of replications = 51

Number of converged replications = 51

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Realistic Scenarios](#)

5.4 Comparison

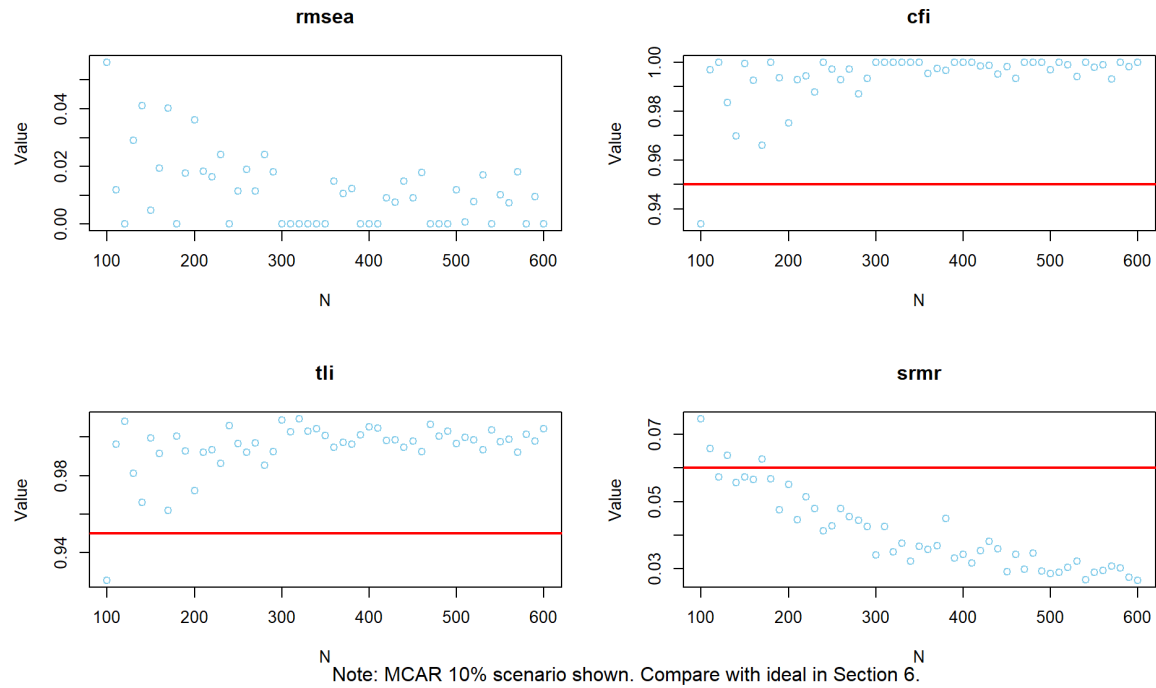


Figure 4: Power curves across population model scenarios (fit-index based)

Source: [Simulation — Realistic Scenarios](#)

Table 5: Required N for 80% power across all analysis scenarios

Scenario	Approach	N for 80% Power	Key Assumption
Analytical (semPower, RMSEA .05)	Analytical	See Section ??	Continuous, normal, ML
Simulation — Ideal (popModel, ML)	Monte Carlo	From Section 6	Continuous, normal, ML
Simulation — Missing data (popModel, 10% MCAR)	Monte Carlo	From sim-missing above	Continuous, normal, ML + 10% MCAR
Simulation — Naive model (loadings = .5)	Monte Carlo	From sim-naive above	Weak loadings (0.5)
Simulation — Optimistic model (loadings = .7)	Monte Carlo	From sim-opt above	Strong loadings (0.7), mod. correlations (0.5)

6 Population Model vs. Empirical Model

Consider an epistemological question driving CFA: can we statistically distinguish the theorized population structure from an empirically estimated model with trivial incidental misspecifications that are ubiquitous in real-life sample covariance matrices? In this section, we compare empirical power distributions generated under a “Null” (empirical sample parameters representing noise) and an “Alternative” (our clean meta-analytic population model).

Null model simulation summary:RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

Alpha						
Fit Indices	0.1	0.05	0.01	0.001	Mean	SD
chisq	276.182	281.907	303.744	310.789	253.064	22.657
aic	59720.961	59758.157	59849.389	59879.379	59439.272	212.914
bic	59993.413	60030.609	60121.842	60151.831	59711.725	212.914
rmsea	0.011	0.012	0.015	0.016	0.006	0.005
cfi	0.996	0.996	0.993	0.992	0.998	0.002
tli	0.996	0.995	0.992	0.991	0.999	0.003
srmr	0.022	0.022	0.023	0.023	0.020	0.001

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5	0.523	0.022	0.024	
psycho=~Q6	0.630	0.031	0.028	
psycho=~Q7	0.548	0.025	0.025	
psycho=~Q11	0.598	0.033	0.029	
psycho=~Q19	0.778	0.022	0.025	
psycho=~Q26	0.543	0.024	0.027	
physical=~Q3	0.415	0.037	0.029	
physical=~Q4	0.387	0.032	0.031	
physical=~Q10	0.670	0.025	0.025	
physical=~Q15	0.388	0.018	0.024	
physical=~Q16	0.552	0.030	0.031	
physical=~Q17	0.812	0.024	0.023	
physical=~Q18	0.812	0.022	0.024	
social=~Q20	0.703	0.023	0.027	
social=~Q21	0.500	0.035	0.033	
social=~Q22	0.634	0.024	0.027	

environment=~Q8	0.445	0.033	0.031			
environment=~Q9	0.512	0.024	0.028			
environment=~Q12	0.750	0.034	0.033			
environment=~Q13	0.497	0.023	0.025			
environment=~Q14	0.648	0.027	0.028			
environment=~Q23	0.493	0.025	0.030			
environment=~Q24	0.469	0.030	0.031			
environment=~Q25	0.467	0.031	0.031			
psycho~~physical	0.897	0.012	0.012			
psycho~~social	0.755	0.020	0.023			
psycho~~environment	0.703	0.026	0.023			
physical~~social	0.615	0.021	0.027			
physical~~environment	0.584	0.029	0.026			
social~~environment	0.553	0.031	0.031			
Q3~~Q4	0.339	0.029	0.027			
Q5~~Q5	0.399	0.019	0.019			
Q6~~Q6	0.579	0.030	0.027			
Q7~~Q7	0.437	0.016	0.021			
Q11~~Q11	0.625	0.023	0.029			
Q19~~Q19	0.291	0.016	0.017			
Q26~~Q26	0.560	0.022	0.026			
Q3~~Q3	0.702	0.034	0.031			
Q4~~Q4	0.862	0.035	0.038			
Q10~~Q10	0.377	0.016	0.019			
Q15~~Q15	0.472	0.023	0.021			
Q16~~Q16	0.755	0.036	0.034			
Q17~~Q17	0.192	0.012	0.013			
Q18~~Q18	0.255	0.015	0.015			
Q20~~Q20	0.301	0.024	0.024			
Q21~~Q21	0.764	0.035	0.036			
Q22~~Q22	0.378	0.023	0.023			
Q8~~Q8	0.756	0.032	0.035			
Q9~~Q9	0.550	0.029	0.027			
Q12~~Q12	0.647	0.036	0.035			
Q13~~Q13	0.432	0.026	0.021			
Q14~~Q14	0.492	0.025	0.026			
Q23~~Q23	0.671	0.034	0.032			
Q24~~Q24	0.714	0.033	0.033			
Q25~~Q25	0.726	0.030	0.034			
	Power (Not equal 0)	Std Est	Std Est	SD	Std Ave	SE
psycho=~Q5		1	0.637	0.016		0.020
psycho=~Q6		1	0.638	0.022		0.020
psycho=~Q7		1	0.638	0.020		0.020

psycho=~Q11	1	0.603	0.023	0.021
psycho=~Q19	1	0.822	0.011	0.012
psycho=~Q26	1	0.587	0.020	0.022
physical=~Q3	1	0.443	0.034	0.026
physical=~Q4	1	0.384	0.029	0.028
physical=~Q10	1	0.737	0.015	0.016
physical=~Q15	1	0.491	0.017	0.025
physical=~Q16	1	0.536	0.024	0.024
physical=~Q17	1	0.880	0.009	0.010
physical=~Q18	1	0.849	0.010	0.011
social=~Q20	1	0.789	0.019	0.019
social=~Q21	1	0.496	0.028	0.027
social=~Q22	1	0.718	0.019	0.021
environment=~Q8	1	0.455	0.028	0.028
environment=~Q9	1	0.568	0.023	0.024
environment=~Q12	1	0.681	0.022	0.021
environment=~Q13	1	0.603	0.022	0.023
environment=~Q14	1	0.678	0.020	0.021
environment=~Q23	1	0.516	0.023	0.026
environment=~Q24	1	0.485	0.026	0.027
environment=~Q25	1	0.480	0.024	0.027
psycho~~physical	1	0.897	0.012	0.012
psycho~~social	1	0.755	0.020	0.023
psycho~~environment	1	0.703	0.026	0.023
physical~~social	1	0.615	0.021	0.027
physical~~environment	1	0.584	0.029	0.026
social~~environment	1	0.553	0.031	0.031
Q3~~Q4	1	0.436	0.028	0.025
Q5~~Q5	1	0.593	0.021	0.026
Q6~~Q6	1	0.593	0.028	0.026
Q7~~Q7	1	0.593	0.025	0.026
Q11~~Q11	1	0.636	0.028	0.026
Q19~~Q19	1	0.325	0.018	0.020
Q26~~Q26	1	0.655	0.023	0.026
Q3~~Q3	1	0.802	0.030	0.023
Q4~~Q4	1	0.851	0.022	0.021
Q10~~Q10	1	0.457	0.022	0.023
Q15~~Q15	1	0.758	0.017	0.025
Q16~~Q16	1	0.712	0.025	0.025
Q17~~Q17	1	0.226	0.015	0.017
Q18~~Q18	1	0.279	0.016	0.018
Q20~~Q20	1	0.378	0.029	0.031
Q21~~Q21	1	0.753	0.027	0.027

Q22~~Q22	1	0.485	0.028	0.030
Q8~~Q8	1	0.792	0.026	0.025
Q9~~Q9	1	0.677	0.026	0.028
Q12~~Q12	1	0.535	0.030	0.028
Q13~~Q13	1	0.636	0.027	0.028
Q14~~Q14	1	0.539	0.027	0.028
Q23~~Q23	1	0.734	0.024	0.027
Q24~~Q24	1	0.764	0.026	0.026
Q25~~Q25	1	0.769	0.023	0.026

	Average Param	Average Bias	Coverage
psycho==~Q5	0.529	-0.006	0.96
psycho==~Q6	0.629	0.001	0.94
psycho==~Q7	0.546	0.002	0.96
psycho==~Q11	0.597	0.001	0.92
psycho==~Q19	0.778	0.000	0.98
psycho==~Q26	0.543	0.001	0.98
physical==~Q3	0.414	0.002	0.86
physical==~Q4	0.387	0.000	0.90
physical==~Q10	0.671	-0.002	0.94
physical==~Q15	0.390	-0.003	1.00
physical==~Q16	0.552	0.000	0.94
physical==~Q17	0.812	0.000	0.90
physical==~Q18	0.813	-0.001	0.96
social==~Q20	0.697	0.007	0.96
social==~Q21	0.500	0.000	0.92
social==~Q22	0.633	0.000	0.96
environment==~Q8	0.450	-0.005	0.96
environment==~Q9	0.513	-0.001	0.96
environment==~Q12	0.740	0.010	0.94
environment==~Q13	0.501	-0.004	0.94
environment==~Q14	0.653	-0.005	0.96
environment==~Q23	0.504	-0.011	0.96
environment==~Q24	0.471	-0.002	0.94
environment==~Q25	0.473	-0.006	0.96
psycho~~physical	0.897	0.000	0.96
psycho~~social	0.762	-0.007	0.96
psycho~~environment	0.702	0.001	0.96
physical~~social	0.620	-0.005	1.00
physical~~environment	0.581	0.003	0.94
social~~environment	0.552	0.001	0.96
Q3~~Q4	0.337	0.002	0.94
Q5~~Q5	0.397	0.002	0.96
Q6~~Q6	0.578	0.002	0.92

Q7~~Q7	0.442	-0.005	0.98
Q11~~Q11	0.626	-0.001	0.98
Q19~~Q19	0.293	-0.001	0.96
Q26~~Q26	0.559	0.001	0.98
Q3~~Q3	0.700	0.002	0.94
Q4~~Q4	0.861	0.001	0.98
Q10~~Q10	0.379	-0.002	0.94
Q15~~Q15	0.475	-0.003	0.94
Q16~~Q16	0.765	-0.010	0.92
Q17~~Q17	0.191	0.001	0.98
Q18~~Q18	0.253	0.001	0.98
Q20~~Q20	0.305	-0.004	0.94
Q21~~Q21	0.757	0.007	0.98
Q22~~Q22	0.378	0.000	0.94
Q8~~Q8	0.755	0.001	1.00
Q9~~Q9	0.556	-0.006	0.90
Q12~~Q12	0.652	-0.004	0.94
Q13~~Q13	0.428	0.004	0.86
Q14~~Q14	0.493	0.000	0.98
Q23~~Q23	0.677	-0.006	0.94
Q24~~Q24	0.717	-0.002	0.98
Q25~~Q25	0.727	-0.001	0.98

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr
chisq	1.000	-0.158	-0.158	0.949	-0.903	-0.999	0.774
aic	-0.158	1.000	1.000	-0.177	0.148	0.160	0.022
bic	-0.158	1.000	1.000	-0.177	0.148	0.160	0.022
rmsea	0.949	-0.177	-0.177	1.000	-0.939	-0.949	0.728
cfi	-0.903	0.148	0.148	-0.939	1.000	0.909	-0.640
tli	-0.999	0.160	0.160	-0.949	0.909	1.000	-0.774
srmr	0.774	0.022	0.022	0.728	-0.640	-0.774	1.000

===== Replications =====

Number of replications = 50

Number of converged replications = 50

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

Source: [Model Comparison \(Null vs Alternative\)](#)

Alternative model simulation summary:RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

Alpha						
Fit Indices	0.1	0.05	0.01	0.001	Mean	SD
chisq	275.814	285.328	303.090	312.391	251.843	22.549
aic	57453.761	57485.469	57570.502	57599.395	57166.971	211.410
bic	57726.214	57757.922	57842.954	57871.847	57439.423	211.410
rmsea	0.011	0.013	0.015	0.016	0.005	0.005
cfi	0.998	0.997	0.996	0.995	0.999	0.001
tli	0.998	0.997	0.995	0.995	0.999	0.002
srmr	0.023	0.024	0.024	0.024	0.021	0.002

===== Parameter Estimates and Standard Errors =====

	Estimate	Average	Estimate	SD	Average	SE
psycho=~Q5		0.918		0.025		0.024
psycho=~Q6		0.809		0.024		0.026
psycho=~Q7		0.939		0.025		0.023
psycho=~Q11		0.731		0.023		0.027
psycho=~Q19		0.747		0.026		0.027
psycho=~Q26		0.636		0.030		0.028
physical=~Q3		0.838		0.025		0.026
physical=~Q4		0.828		0.026		0.026
physical=~Q10		0.632		0.031		0.029
physical=~Q15		0.704		0.026		0.027
physical=~Q16		0.679		0.028		0.028
physical=~Q17		0.890		0.023		0.025
physical=~Q18		0.607		0.024		0.029
social=~Q20		0.942		0.023		0.024
social=~Q21		0.892		0.023		0.025
social=~Q22		0.781		0.032		0.027
environment=~Q8		0.368		0.038		0.031
environment=~Q9		0.490		0.031		0.031
environment=~Q12		0.791		0.026		0.027
environment=~Q13		0.810		0.028		0.027
environment=~Q14		0.774		0.029		0.027
environment=~Q23		0.747		0.029		0.027
environment=~Q24		0.644		0.029		0.029
environment=~Q25		0.731		0.029		0.028
psycho~~physical		0.204		0.031		0.032
psycho~~social		0.204		0.034		0.032
psycho~~environment		0.189		0.035		0.032
physical~~social		0.199		0.032		0.032

physical~~environment	0.204	0.036	0.033
social~~environment	0.199	0.035	0.033
Q3~~Q4	0.091	0.013	0.014
Q5~~Q5	0.155	0.010	0.010
Q6~~Q6	0.344	0.016	0.017
Q7~~Q7	0.116	0.010	0.009
Q11~~Q11	0.468	0.021	0.022
Q19~~Q19	0.436	0.023	0.021
Q26~~Q26	0.599	0.025	0.027
Q3~~Q3	0.294	0.017	0.018
Q4~~Q4	0.310	0.016	0.018
Q10~~Q10	0.600	0.027	0.028
Q15~~Q15	0.488	0.024	0.024
Q16~~Q16	0.538	0.027	0.026
Q17~~Q17	0.209	0.016	0.015
Q18~~Q18	0.627	0.025	0.029
Q20~~Q20	0.116	0.016	0.015
Q21~~Q21	0.211	0.015	0.016
Q22~~Q22	0.389	0.018	0.020
Q8~~Q8	0.863	0.032	0.038
Q9~~Q9	0.763	0.032	0.035
Q12~~Q12	0.370	0.020	0.020
Q13~~Q13	0.349	0.024	0.020
Q14~~Q14	0.398	0.024	0.021
Q23~~Q23	0.443	0.028	0.023
Q24~~Q24	0.590	0.023	0.028
Q25~~Q25	0.468	0.024	0.024

	Power (Not equal 0)	Std Est	Std Est	SD	Std Ave SE
psycho=~Q5	1	0.919	0.006		0.006
psycho=~Q6	1	0.809	0.011		0.012
psycho=~Q7	1	0.940	0.006		0.005
psycho=~Q11	1	0.730	0.014		0.015
psycho=~Q19	1	0.749	0.013		0.015
psycho=~Q26	1	0.634	0.020		0.019
physical=~Q3	1	0.839	0.011		0.012
physical=~Q4	1	0.830	0.011		0.012
physical=~Q10	1	0.632	0.021		0.020
physical=~Q15	1	0.709	0.018		0.017
physical=~Q16	1	0.679	0.017		0.018
physical=~Q17	1	0.889	0.009		0.009
physical=~Q18	1	0.609	0.017		0.021
social=~Q20	1	0.940	0.009		0.008
social=~Q21	1	0.889	0.009		0.010

social=~Q22	1	0.781	0.016	0.014
environment=~Q8	1	0.367	0.033	0.028
environment=~Q9	1	0.489	0.025	0.025
environment=~Q12	1	0.793	0.014	0.014
environment=~Q13	1	0.808	0.015	0.013
environment=~Q14	1	0.775	0.017	0.015
environment=~Q23	1	0.746	0.020	0.016
environment=~Q24	1	0.643	0.020	0.020
environment=~Q25	1	0.730	0.018	0.017
psycho~~physical	1	0.204	0.031	0.032
psycho~~social	1	0.204	0.034	0.032
psycho~~environment	1	0.189	0.035	0.032
physical~~social	1	0.199	0.032	0.032
physical~~environment	1	0.204	0.036	0.033
social~~environment	1	0.199	0.035	0.033
Q3~~Q4	1	0.300	0.034	0.037
Q5~~Q5	1	0.156	0.012	0.012
Q6~~Q6	1	0.345	0.018	0.019
Q7~~Q7	1	0.116	0.011	0.010
Q11~~Q11	1	0.467	0.021	0.022
Q19~~Q19	1	0.439	0.020	0.022
Q26~~Q26	1	0.597	0.026	0.025
Q3~~Q3	1	0.295	0.019	0.020
Q4~~Q4	1	0.311	0.019	0.020
Q10~~Q10	1	0.601	0.026	0.025
Q15~~Q15	1	0.497	0.025	0.024
Q16~~Q16	1	0.538	0.024	0.025
Q17~~Q17	1	0.209	0.016	0.016
Q18~~Q18	1	0.629	0.020	0.026
Q20~~Q20	1	0.115	0.017	0.016
Q21~~Q21	1	0.210	0.016	0.017
Q22~~Q22	1	0.390	0.025	0.021
Q8~~Q8	1	0.864	0.024	0.021
Q9~~Q9	1	0.760	0.024	0.025
Q12~~Q12	1	0.372	0.023	0.022
Q13~~Q13	1	0.347	0.024	0.021
Q14~~Q14	1	0.400	0.026	0.023
Q23~~Q23	1	0.443	0.029	0.024
Q24~~Q24	1	0.587	0.025	0.026
Q25~~Q25	1	0.467	0.027	0.024
Average Param Average Bias Coverage				
psycho=~Q5	0.920	-0.002	0.90	
psycho=~Q6	0.810	-0.001	0.96	

psycho=~Q7	0.940	-0.001	0.90
psycho=~Q11	0.730	0.001	1.00
psycho=~Q19	0.750	-0.003	0.96
psycho=~Q26	0.630	0.006	0.90
physical=~Q3	0.840	-0.002	0.96
physical=~Q4	0.830	-0.002	0.98
physical=~Q10	0.630	0.002	0.92
physical=~Q15	0.710	-0.006	0.98
physical=~Q16	0.680	-0.001	0.92
physical=~Q17	0.890	0.000	0.98
physical=~Q18	0.610	-0.003	1.00
social=~Q20	0.940	0.002	0.96
social=~Q21	0.890	0.002	0.98
social=~Q22	0.780	0.001	0.88
environment=~Q8	0.370	-0.002	0.90
environment=~Q9	0.490	0.000	0.98
environment=~Q12	0.790	0.001	1.00
environment=~Q13	0.810	0.000	0.98
environment=~Q14	0.770	0.004	0.98
environment=~Q23	0.750	-0.003	0.88
environment=~Q24	0.640	0.004	0.94
environment=~Q25	0.730	0.001	0.96
psycho~~physical	0.200	0.004	0.96
psycho~~social	0.200	0.004	0.90
psycho~~environment	0.200	-0.011	0.90
physical~~social	0.200	-0.001	0.96
physical~~environment	0.200	0.004	0.90
social~~environment	0.200	-0.001	0.94
Q3~~Q4	0.091	0.000	0.98
Q5~~Q5	0.154	0.001	0.96
Q6~~Q6	0.344	0.000	0.92
Q7~~Q7	0.116	-0.001	0.94
Q11~~Q11	0.467	0.001	0.98
Q19~~Q19	0.438	-0.002	0.92
Q26~~Q26	0.603	-0.004	1.00
Q3~~Q3	0.294	0.000	0.92
Q4~~Q4	0.311	-0.001	0.96
Q10~~Q10	0.603	-0.004	0.94
Q15~~Q15	0.496	-0.008	0.94
Q16~~Q16	0.538	0.000	0.96
Q17~~Q17	0.208	0.001	0.96
Q18~~Q18	0.628	-0.001	0.96
Q20~~Q20	0.116	-0.001	0.92

Q21~~Q21	0.208	0.003	0.96
Q22~~Q22	0.392	-0.003	0.96
Q8~~Q8	0.863	0.000	1.00
Q9~~Q9	0.760	0.003	0.96
Q12~~Q12	0.376	-0.006	0.92
Q13~~Q13	0.344	0.005	0.92
Q14~~Q14	0.407	-0.009	0.86
Q23~~Q23	0.438	0.005	0.90
Q24~~Q24	0.590	-0.001	0.96
Q25~~Q25	0.467	0.001	0.92

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr
chisq	1.000	-0.228	-0.228	0.937	-0.899	-1.000	0.701
aic	-0.228	1.000	1.000	-0.183	0.186	0.226	-0.058
bic	-0.228	1.000	1.000	-0.183	0.186	0.226	-0.058
rmsea	0.937	-0.183	-0.183	1.000	-0.942	-0.938	0.667
cfi	-0.899	0.186	0.186	-0.942	1.000	0.901	-0.625
tli	-1.000	0.226	0.226	-0.938	0.901	1.000	-0.697
srmr	0.701	-0.058	-0.058	0.667	-0.625	-0.697	1.000

===== Replications =====

Number of replications = 50

Number of converged replications = 50

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

Source: [Model Comparison \(Null vs Alternative\)](#)

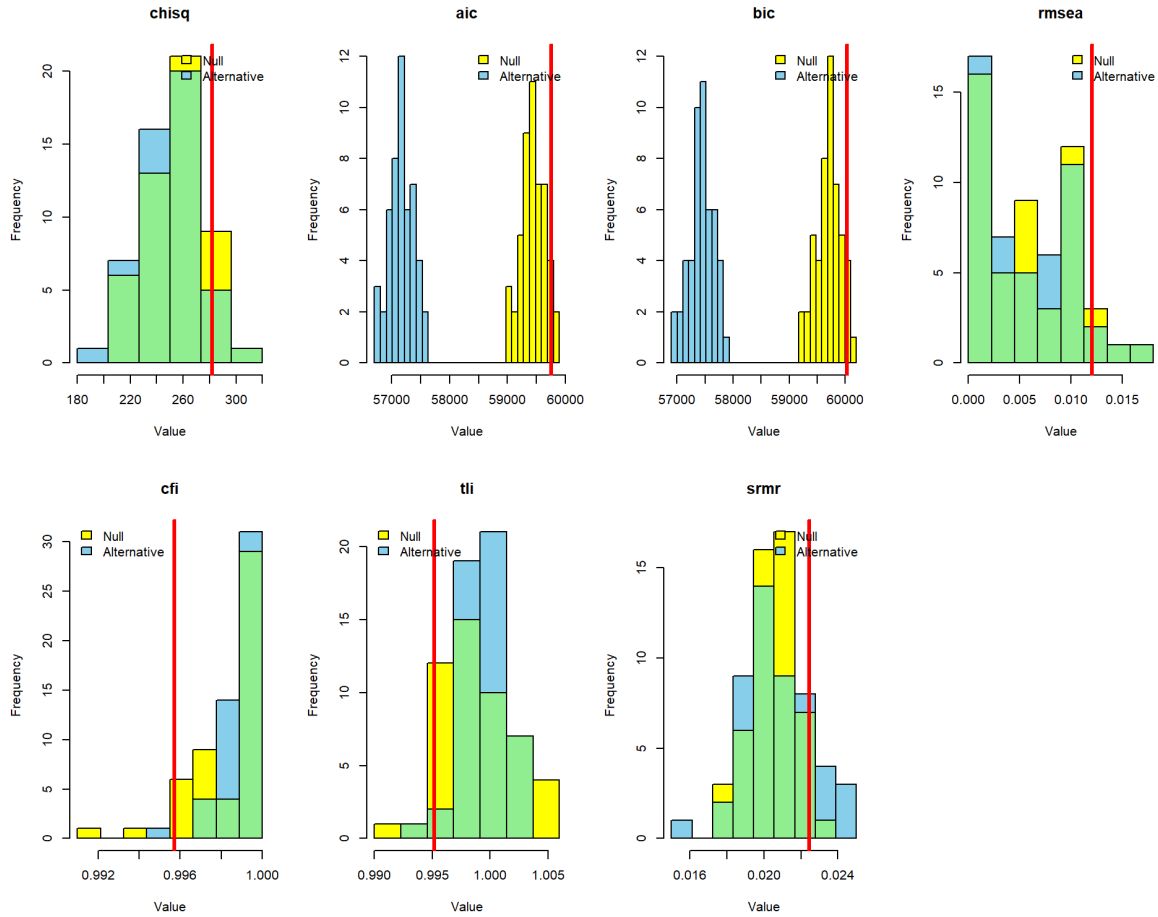


Figure 5: Fit index distributions — Null (empirical) vs. Alternative (population)

Source: [Model Comparison \(Null vs Alternative\)](#)

Table 6: Post hoc power to distinguish empirical from population model (N = 1047)

	Fit Index	Null Cutoff (= .05)	Power
RMSEA	From cutoff_null above		From getPowerFit above
CFI	From cutoff_null above		From getPowerFit above
TLI	From cutoff_null above		From getPowerFit above
SRMR	From cutoff_null above		From getPowerFit above

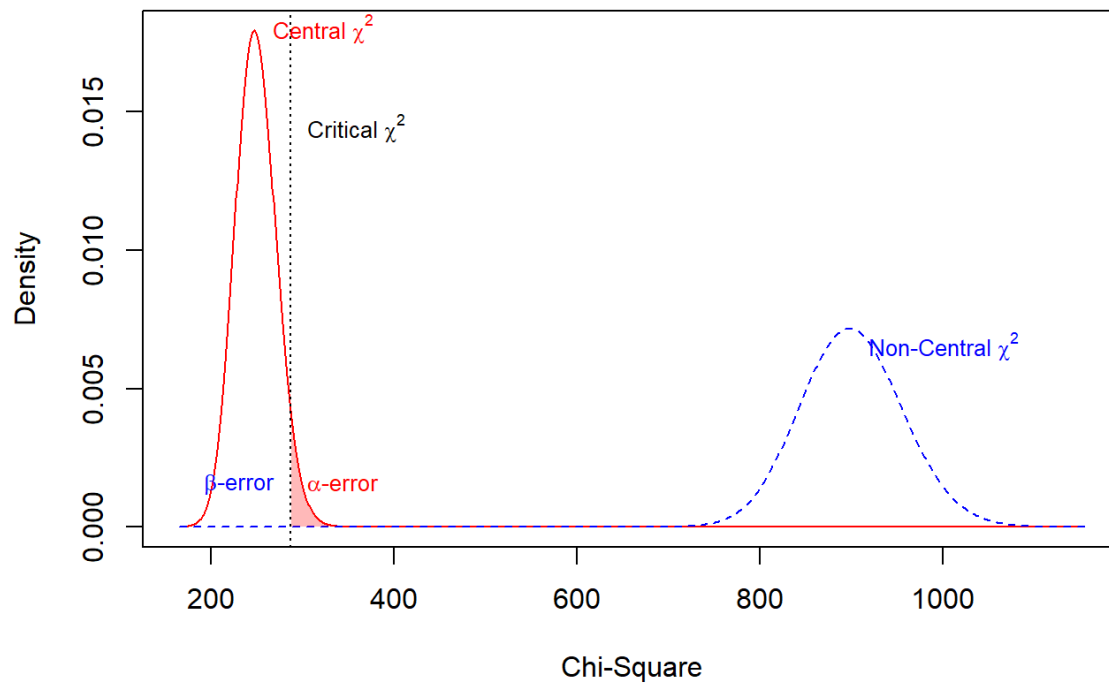
Source: [Model Comparison \(Null vs Alternative\)](#)

7 Post Hoc: Achieved Power

For the empirical WHOQOL-BREF dataset, $N = 1,047$. What is our achieved power if true RMSEA is 0.05?

semPower: Post hoc power analysis

F0	0.622500
RMSEA	0.050000
Mc	0.732531
df	249
Num Observations	1047
NCP	651.1350
Critical Chi-Square	286.8077
Alpha	0.050000
Beta	4.771426e-48
Power (1 - Beta)	> 0.9999
Implied Alpha/Beta Ratio	1.047905e+46



Source: [Post Hoc Power Analysis](#)

Achieved power (`semTools`): 1

Source: [Post Hoc Power Analysis](#)

8 When to Use What: Analytical vs. Simulation

Feature	Analytical (<code>semPower</code> / <code>semTools</code>)	Simulation (<code>simsem</code>)
Speed	Instantaneous	Minutes to hours
Handles non-normality effectively	No	Yes
Handles missing data mechanism	No	Yes

Feature	Analytical (<code>semPower</code> / <code>semTools</code>)	Simulation (<code>simsem</code>)
Parameter-level power targeting	Very Limited	Comprehensive
Coverage / relative bias diagnostics	No	Yes
Primary Use-Case	Grant planning, quick rough parameterization bounds	Pre-registration confirmation, sensitivity checks, final reporting

Best Practice Recommendation: Define theoretical bounds mapping an expected scenario utilizing fast analytical approximations. Validate robust tolerance to observed/anticipated data flaws (e.g., missingness, skew, differing distributions) iteratively through explicit simulation runs before finalizing sample specifications.

References

- Lin, L.-C., & Yao, G. (2022). Validation of the factor structure of the WHOQOL-BREF using meta-analysis of exploratory factor analysis and social network analysis. *Psychological Assessment*, *34*(7), 660–670. <https://doi.org/10.1037/pas0001122>
- MacCallum, R. C., Browne, M. W., & Sugawara, H. M. (1996). Power analysis and determination of sample size for covariance structure modeling. *Psychological Methods*, *1*(2), 130–149. <https://doi.org/10.1037/1082-989X.1.2.130>
- Moshagen, M., & Bader, M. (2024). `semPower`: General Power Analysis for Structural Equation Models. *Behavior Research Methods*, *56*, 2901–2922. <https://doi.org/10.3758/s13428-023-02254-7>
- Mosqueira Taipe, C. M., Amador Campos, J. A., Krieger, V., Miaso, A. I., Molina, N. P. F. M., & dos Santos, P. L. (2026). Systematic review of the factor structure of WHOQOL-bref in adult population. *Quality of Life Research*, *35*(17), 1–18. <https://doi.org/10.1007/s11136-025-04115-6>
- Pornprasertmanit, S., Miller, P., Schoemann, A., & Jorgensen, T. D. (2021). *Simsem: SIMulated Structural Equation Modeling*. <https://CRAN.R-project.org/package=simsem>